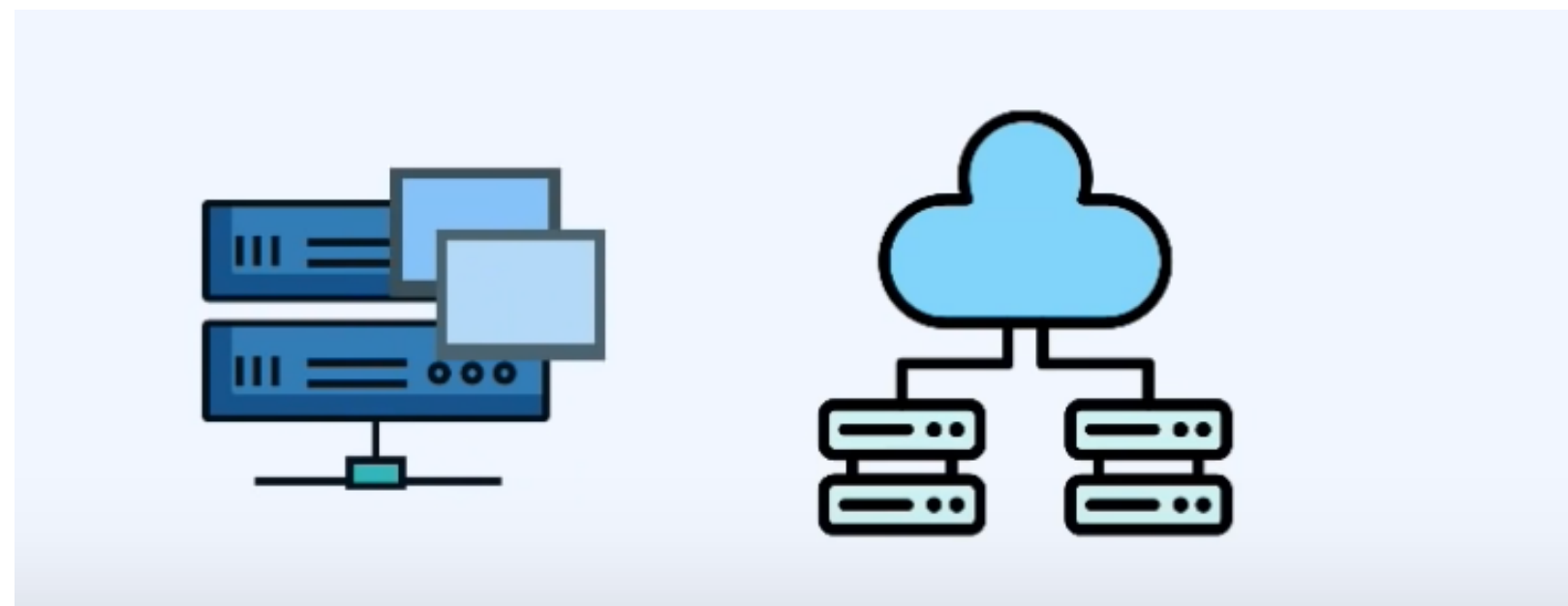


Kubernetes

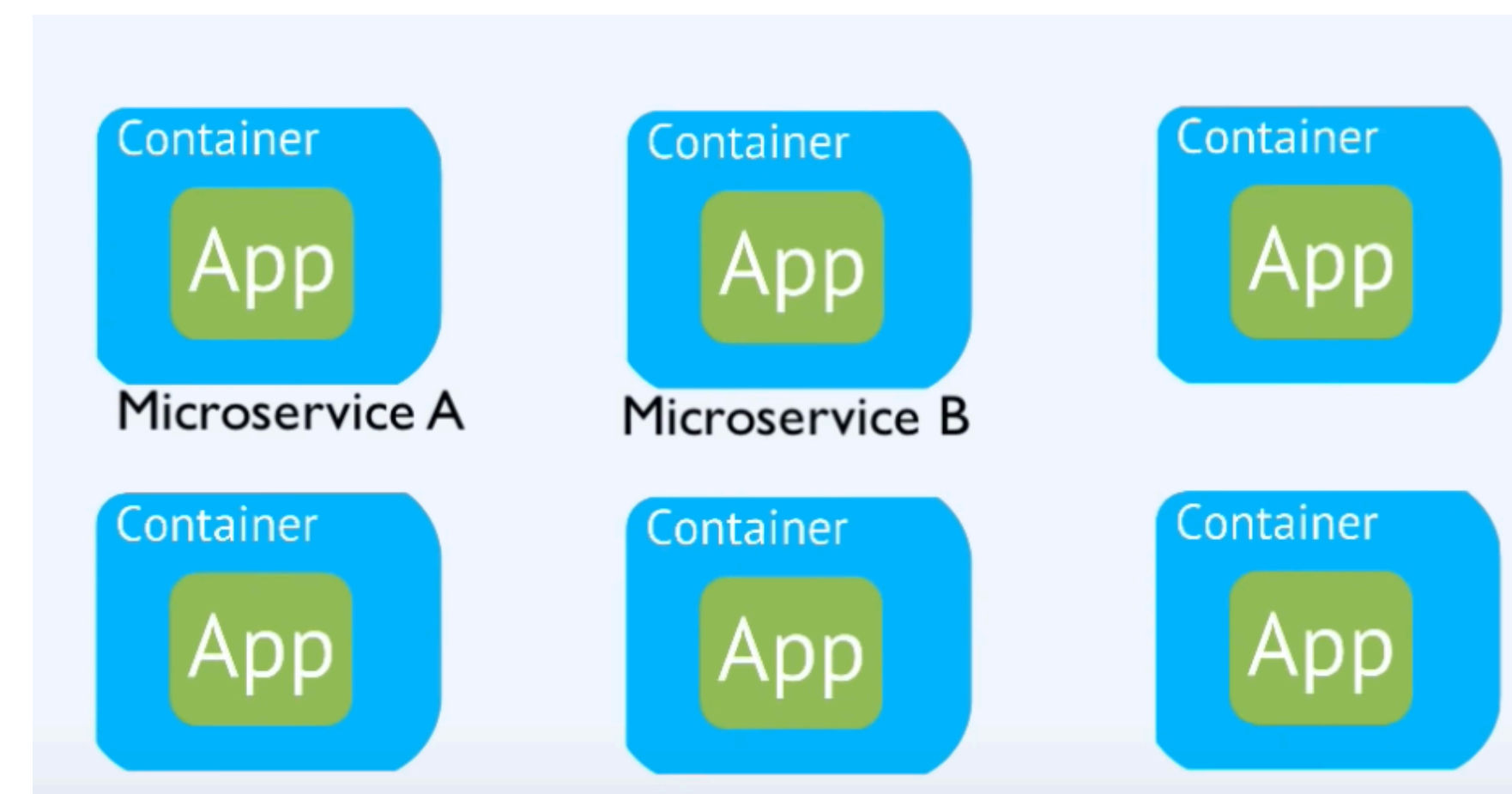
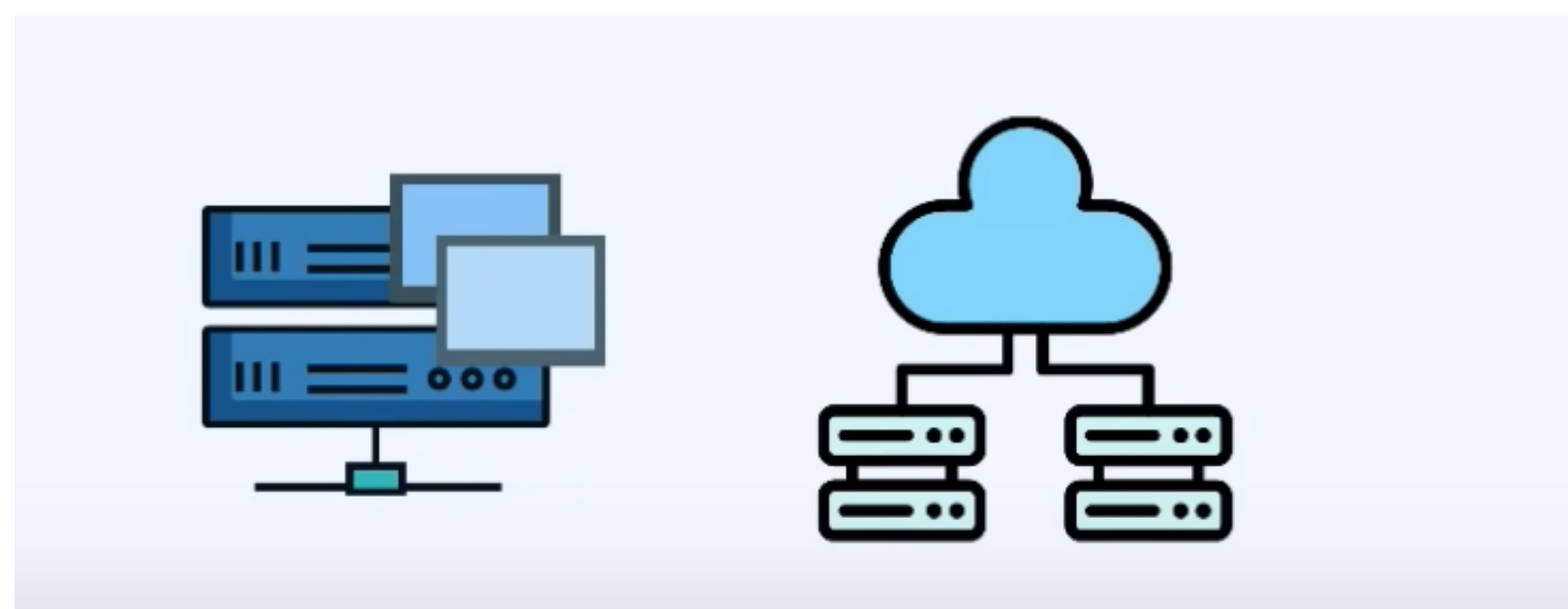
Ideja

- Open source orchestration
- Google
- Administracija vsebnikov



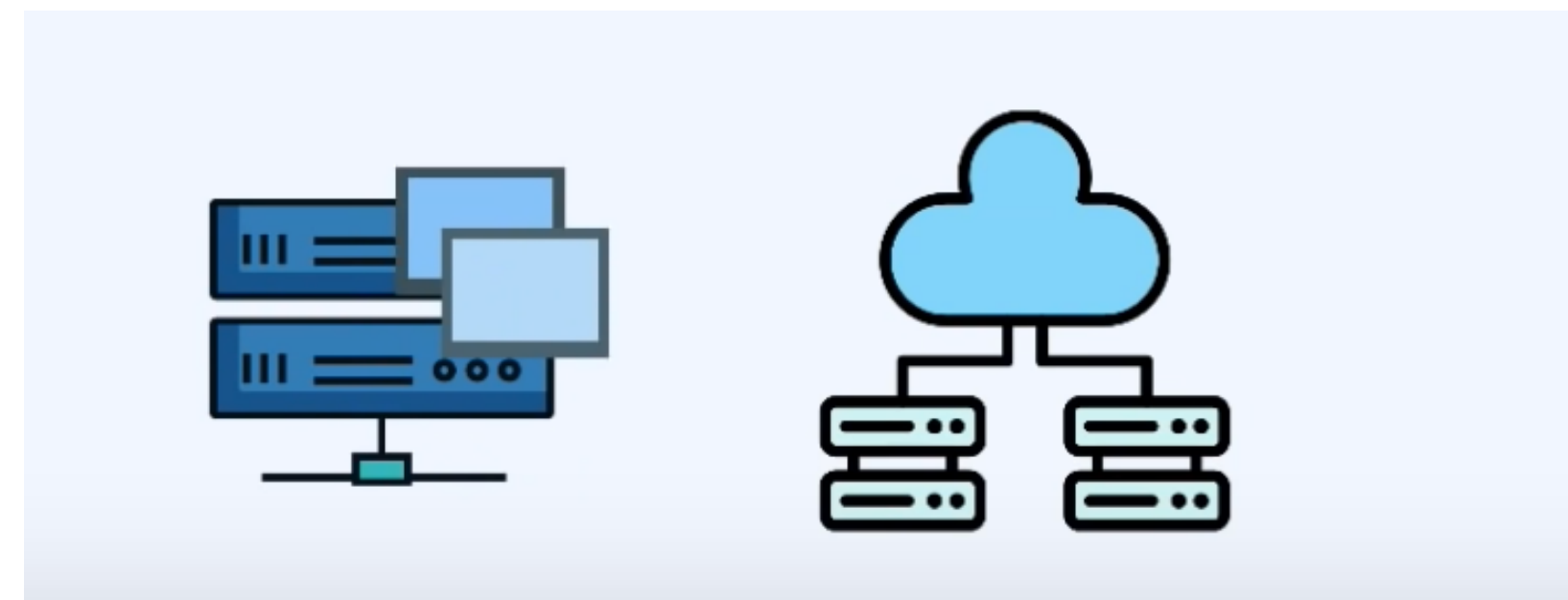
Kaj rešuje?

- Od monolitnih proti Microservices
- Vse pogosta uporaba vsebnikov
- Učinkovit način administracijo na tisočih vsebnikov

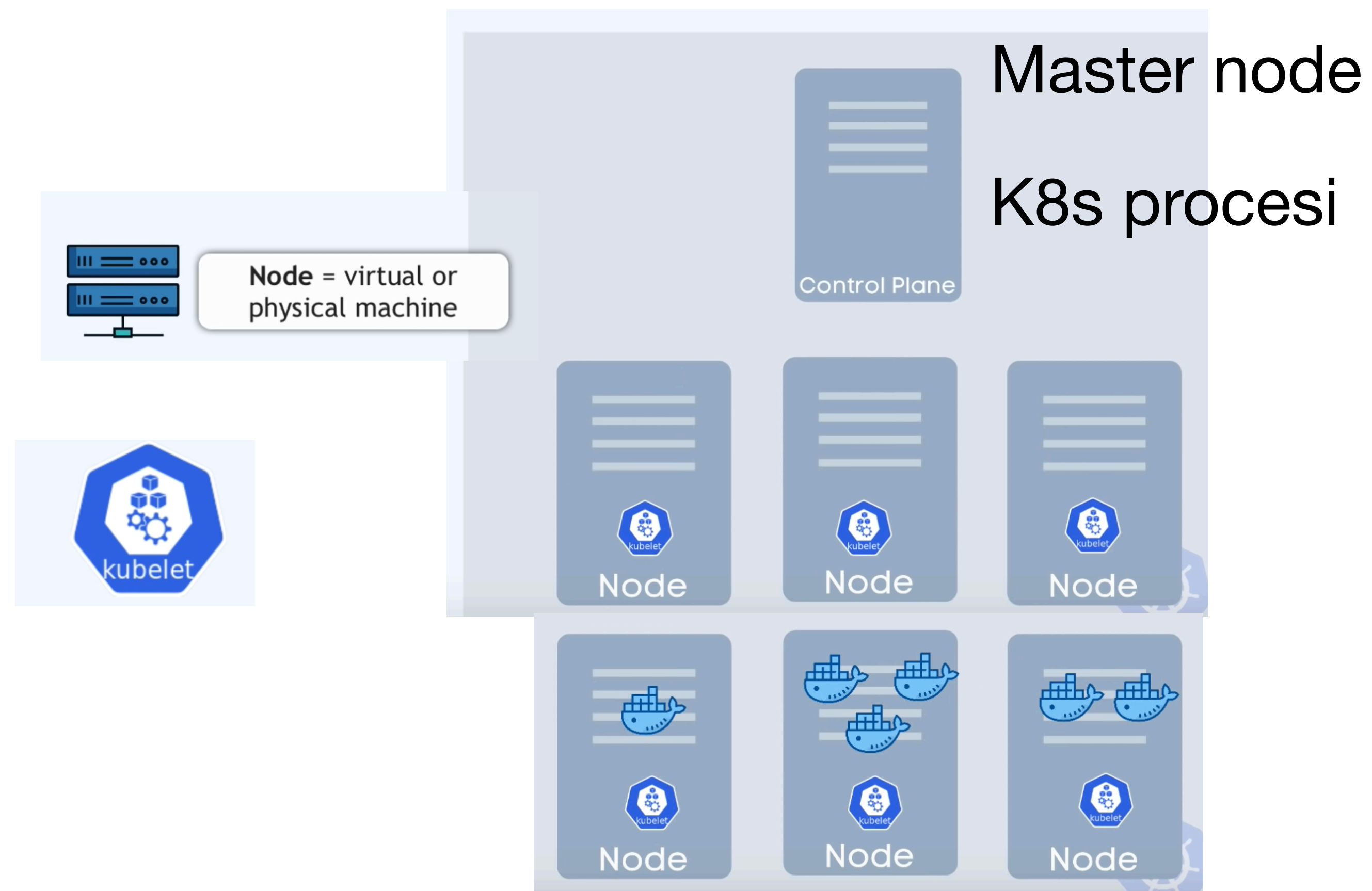


Kaj je orkestracija?

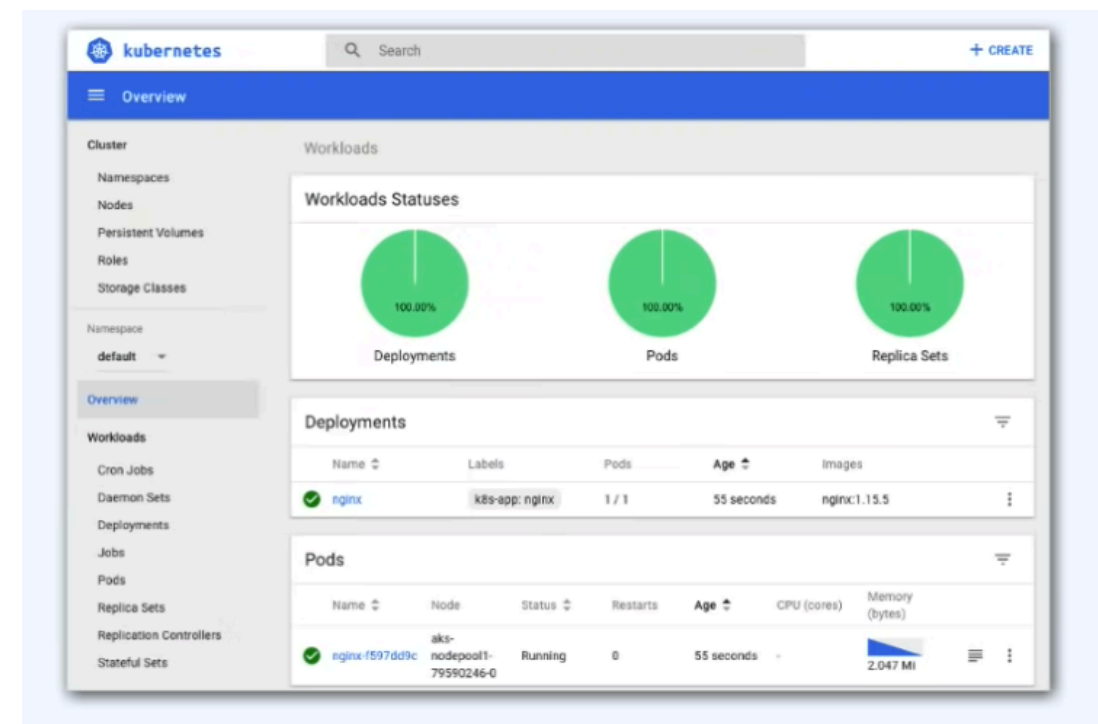
- Visoka dostopnost - low downtime
- Scalability
- Disaster recovery



Arhitektura



Arhitektura



UI

API

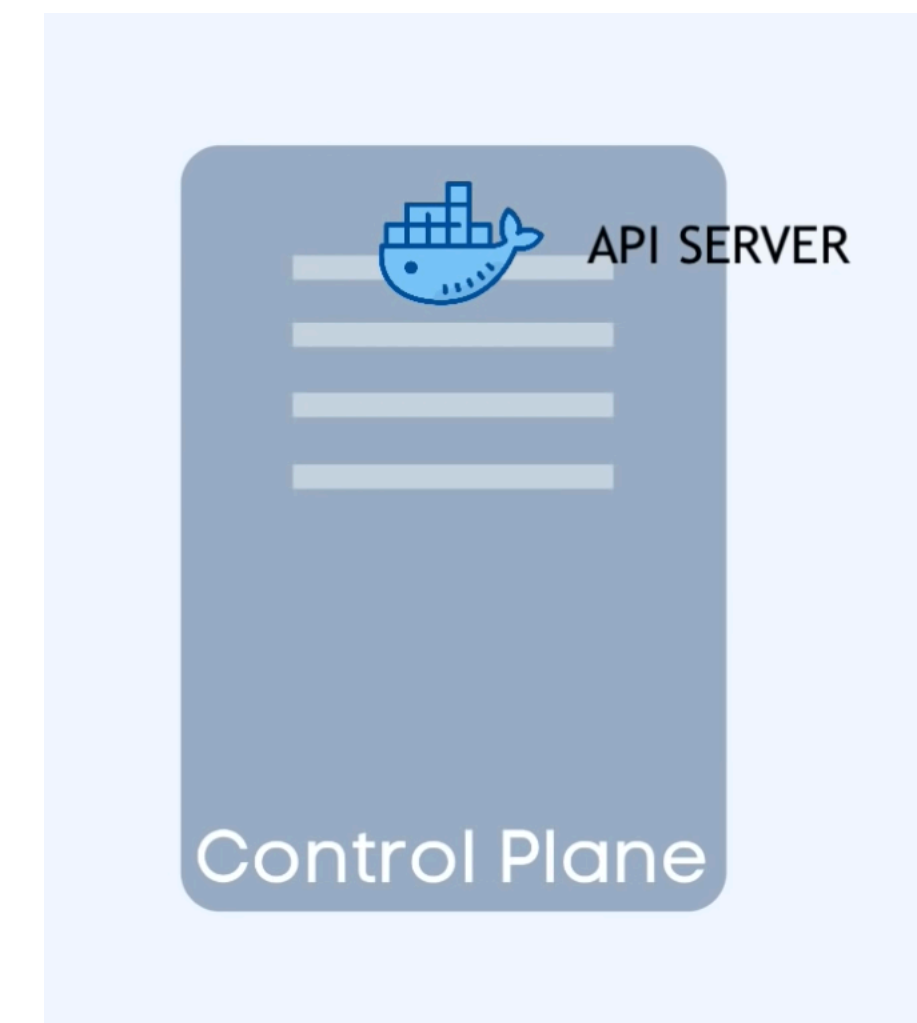
CLI



API Server
Vstopna točka K8

Controller manager
Sledi kaj se vse dogaja

Scheduler
Ureja vsebnike



etcd
DB / storage

Arhitektura

Control plane

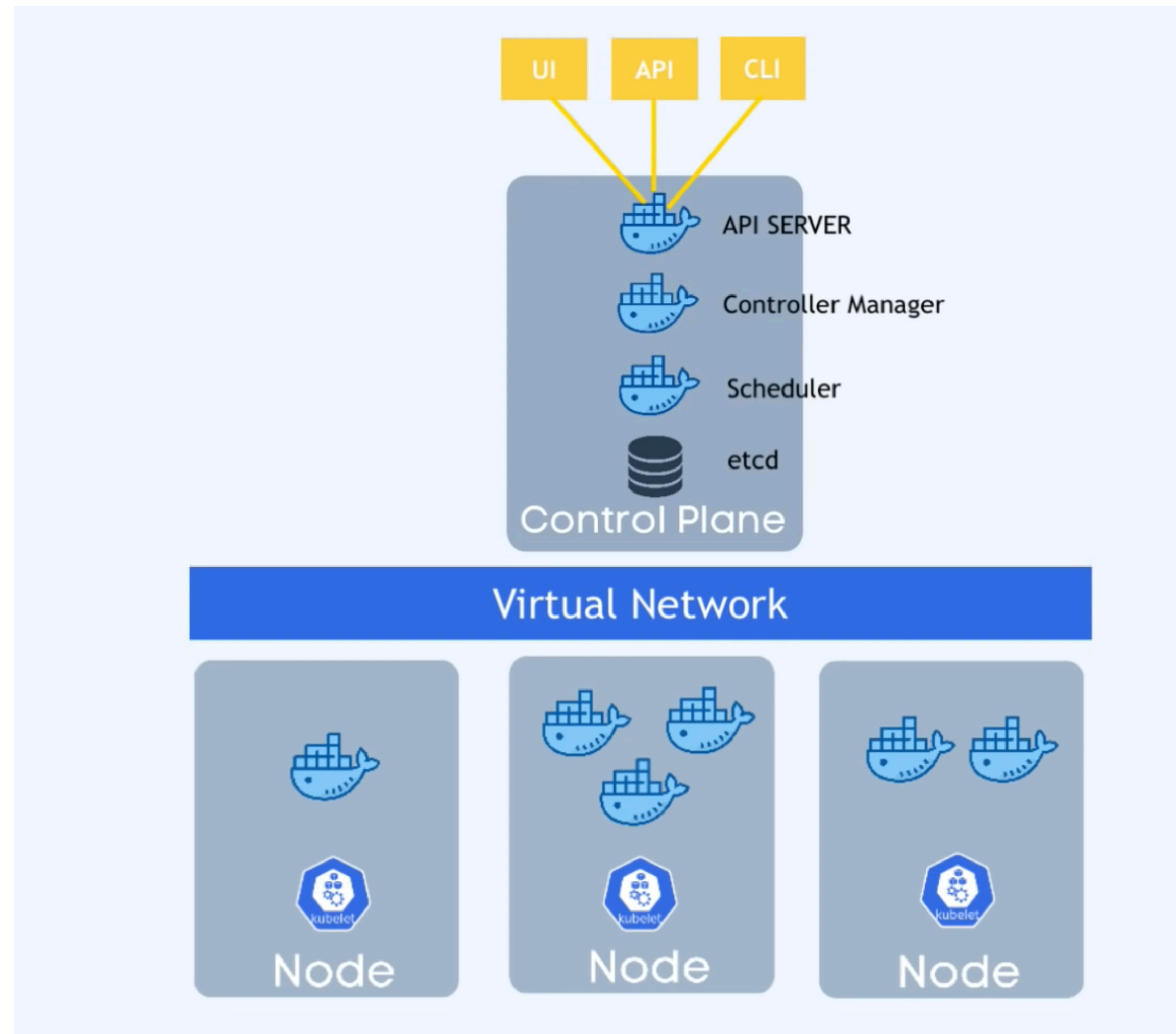
Le peščica procesov

Virtual network

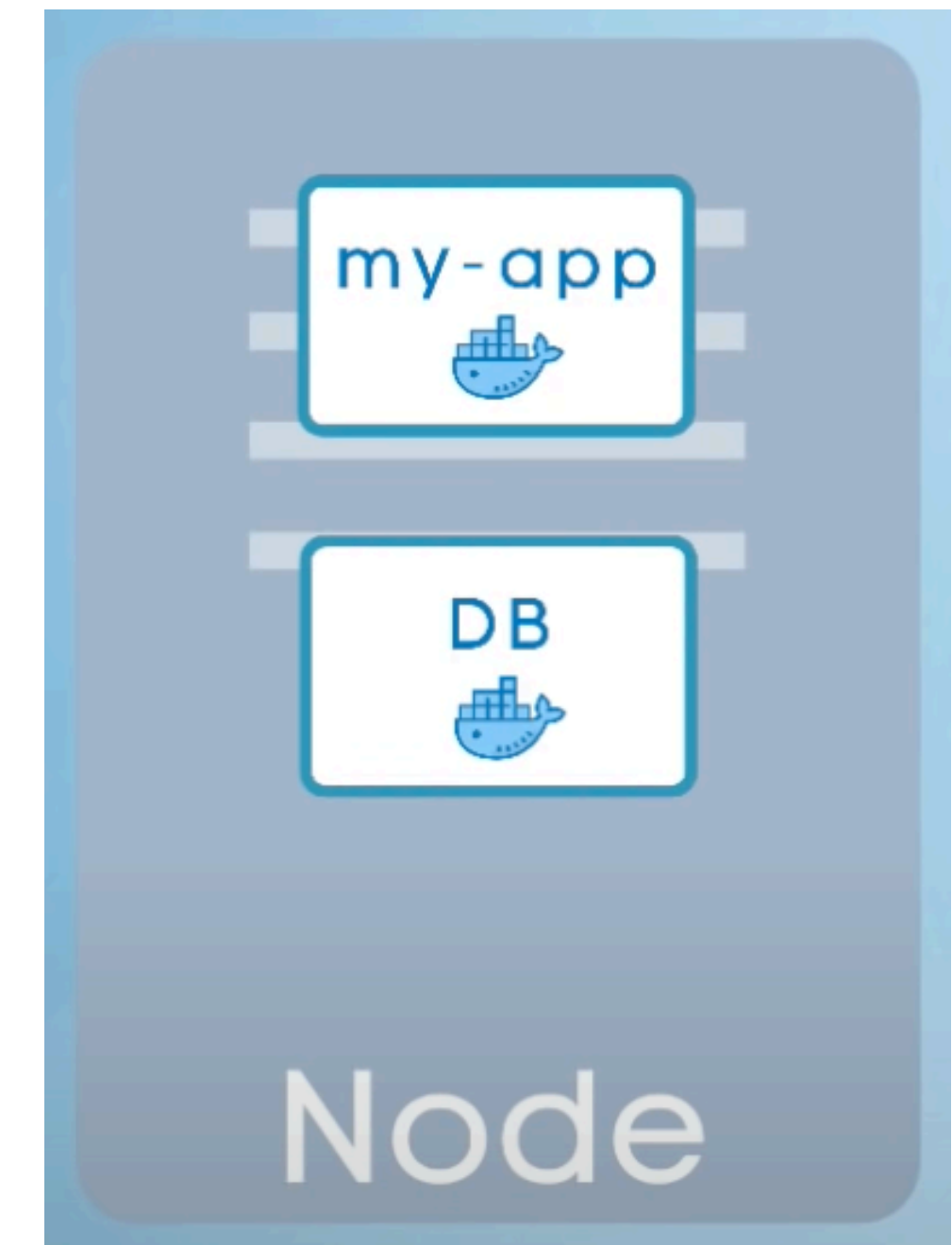
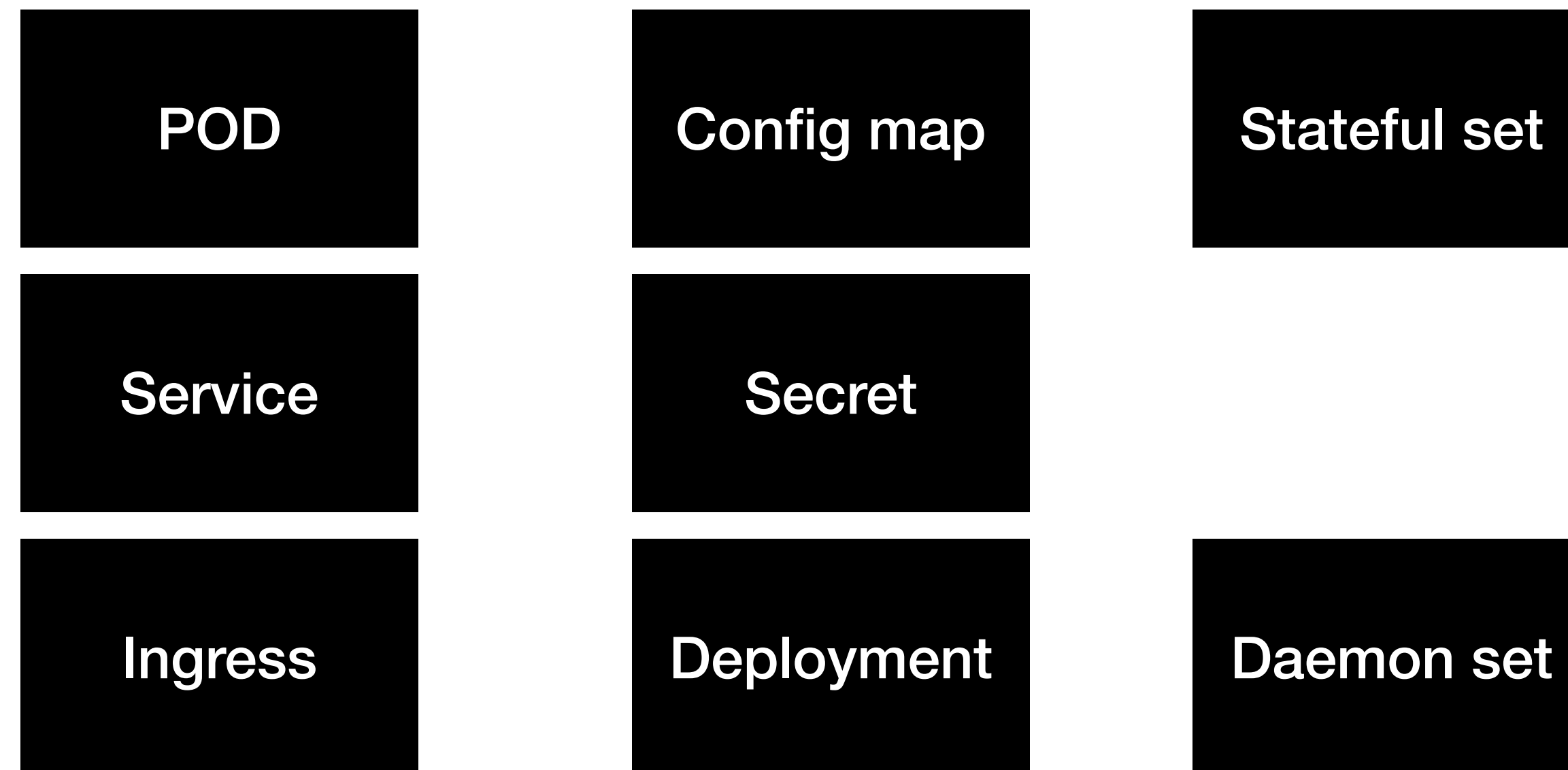
Navidezna poenotena mašina

Worker nodes

Velika računska obremenitev
Več resursov

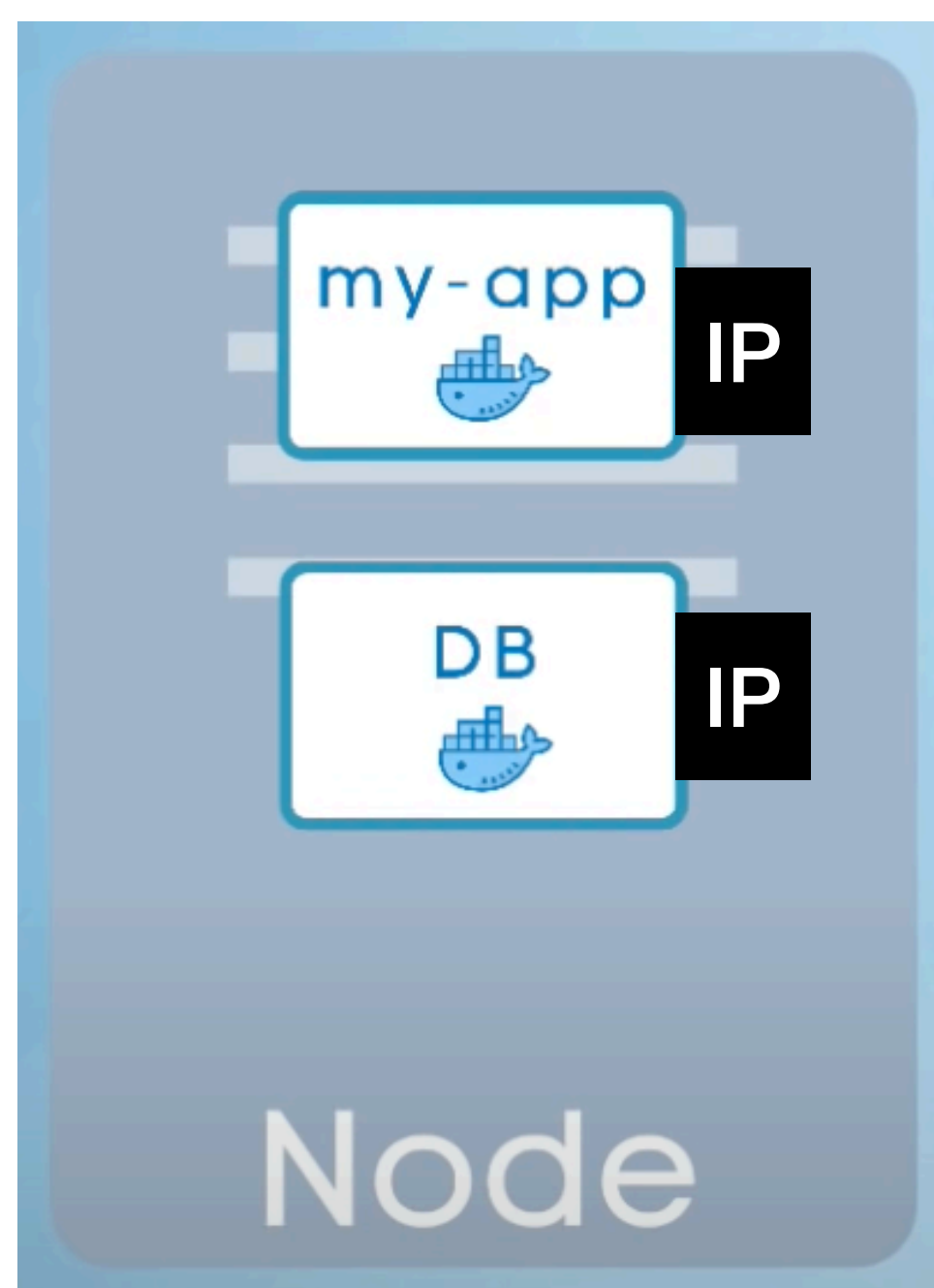


Komponenti



Komponenti

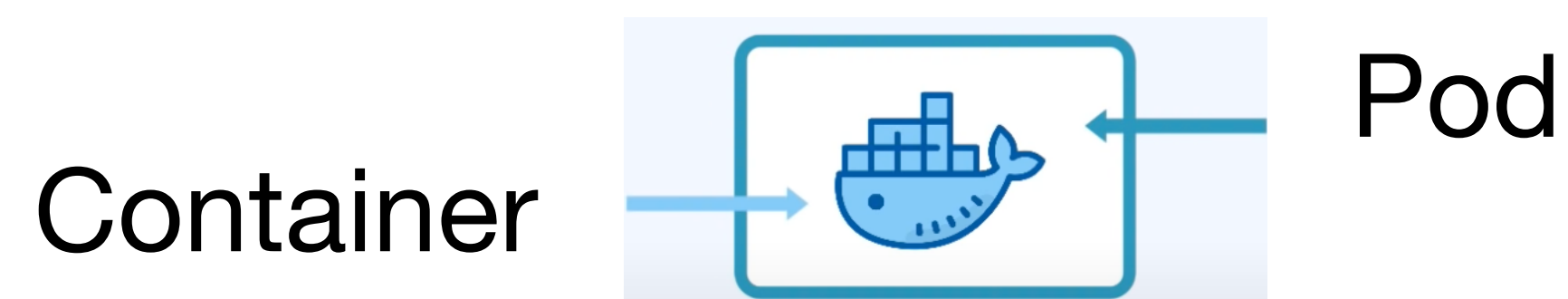
Pod



Node - fizični ali virtualni računalnik



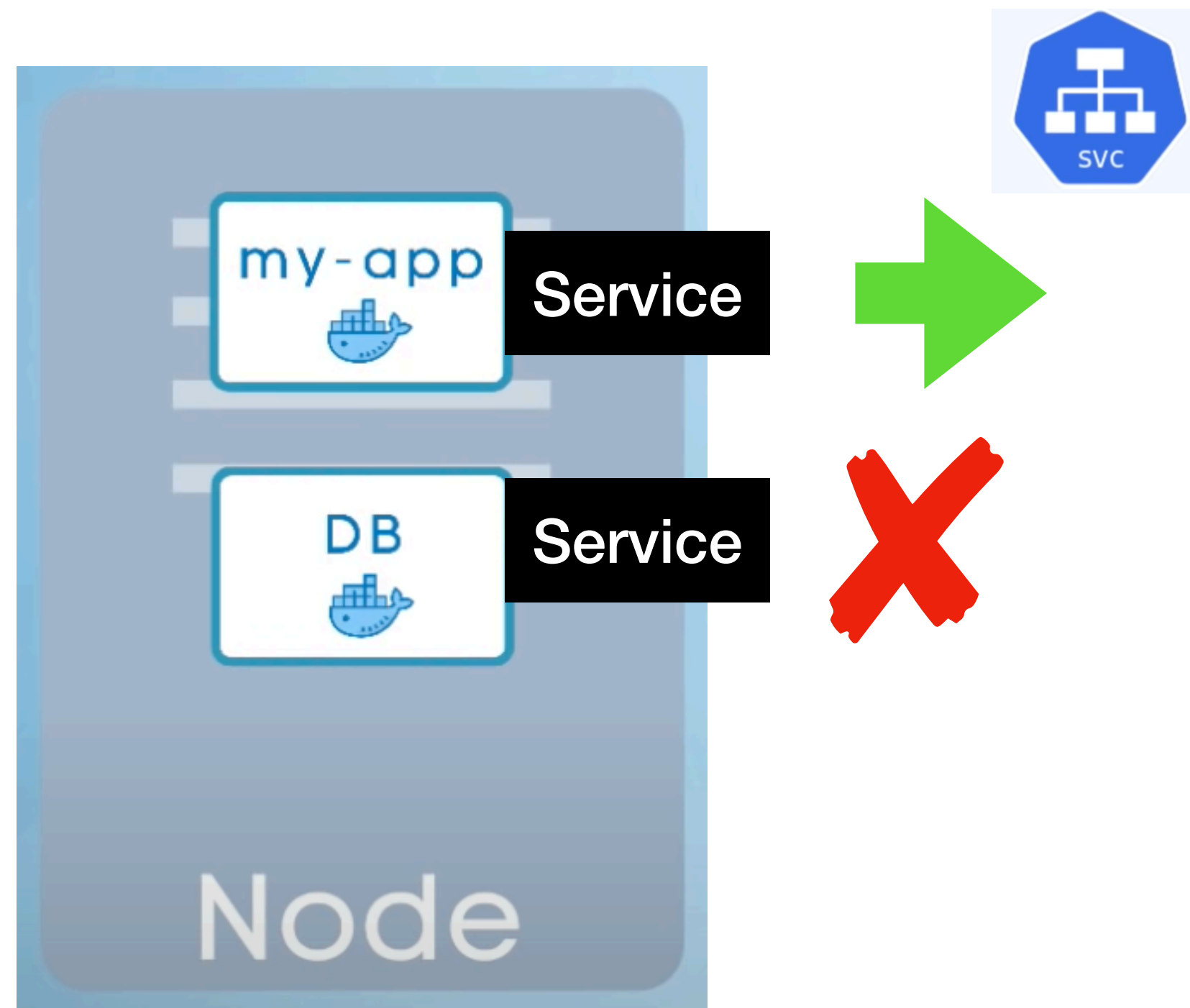
Pod - najmanjša enota



- Običajno 1 aplikacija za 1 pod
- Vsak pod dobi IP
- Pod so *začasni*
- Nov Pod - nov IP

Komponenti

Service



Permanentni IP naslov

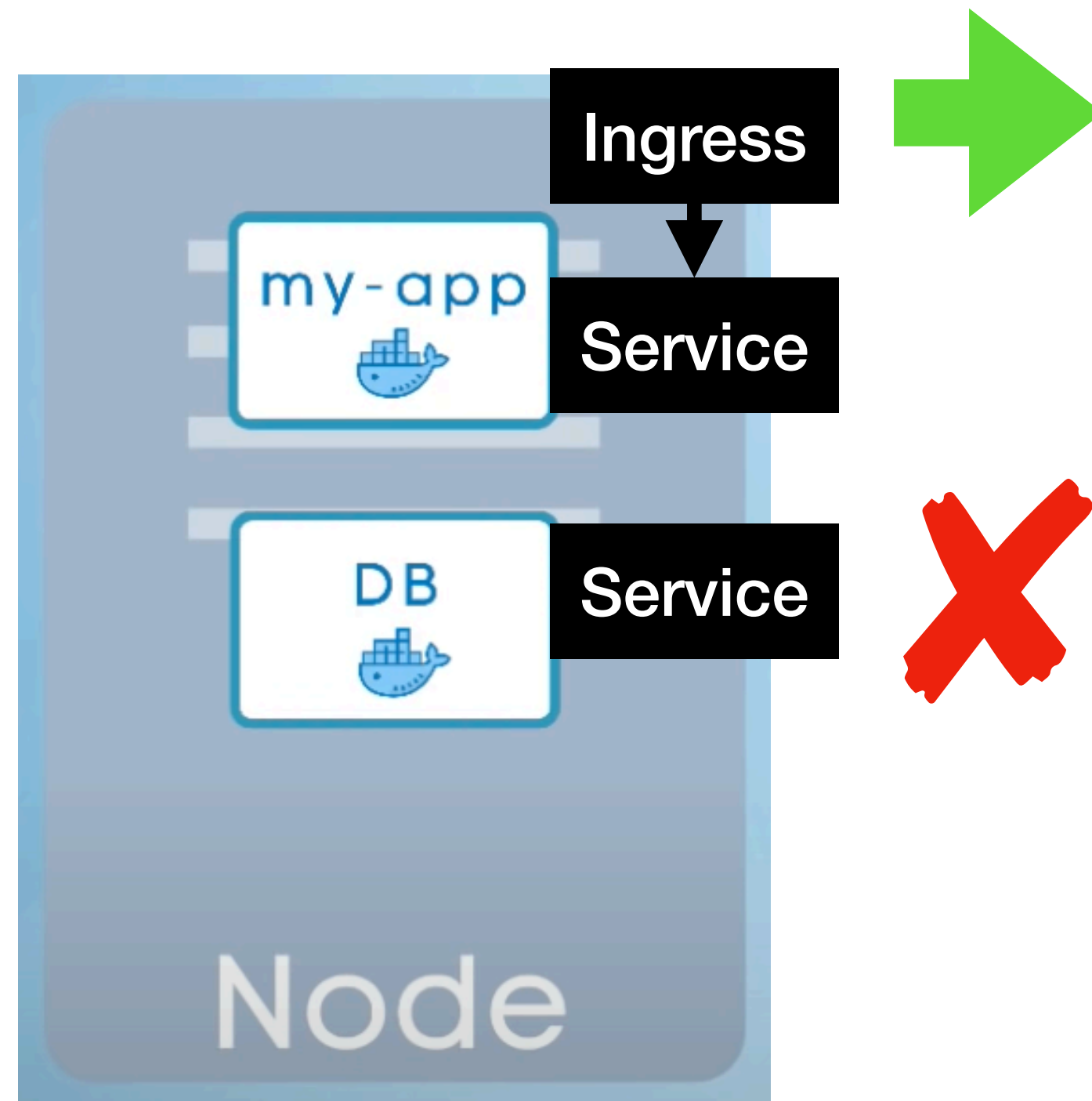
- Pod in service imata različni življenski cikel

Zunanji IP naslov

- External service <http://node ip:port service>
- Privzeto - internal service

Komponenti

Ingress

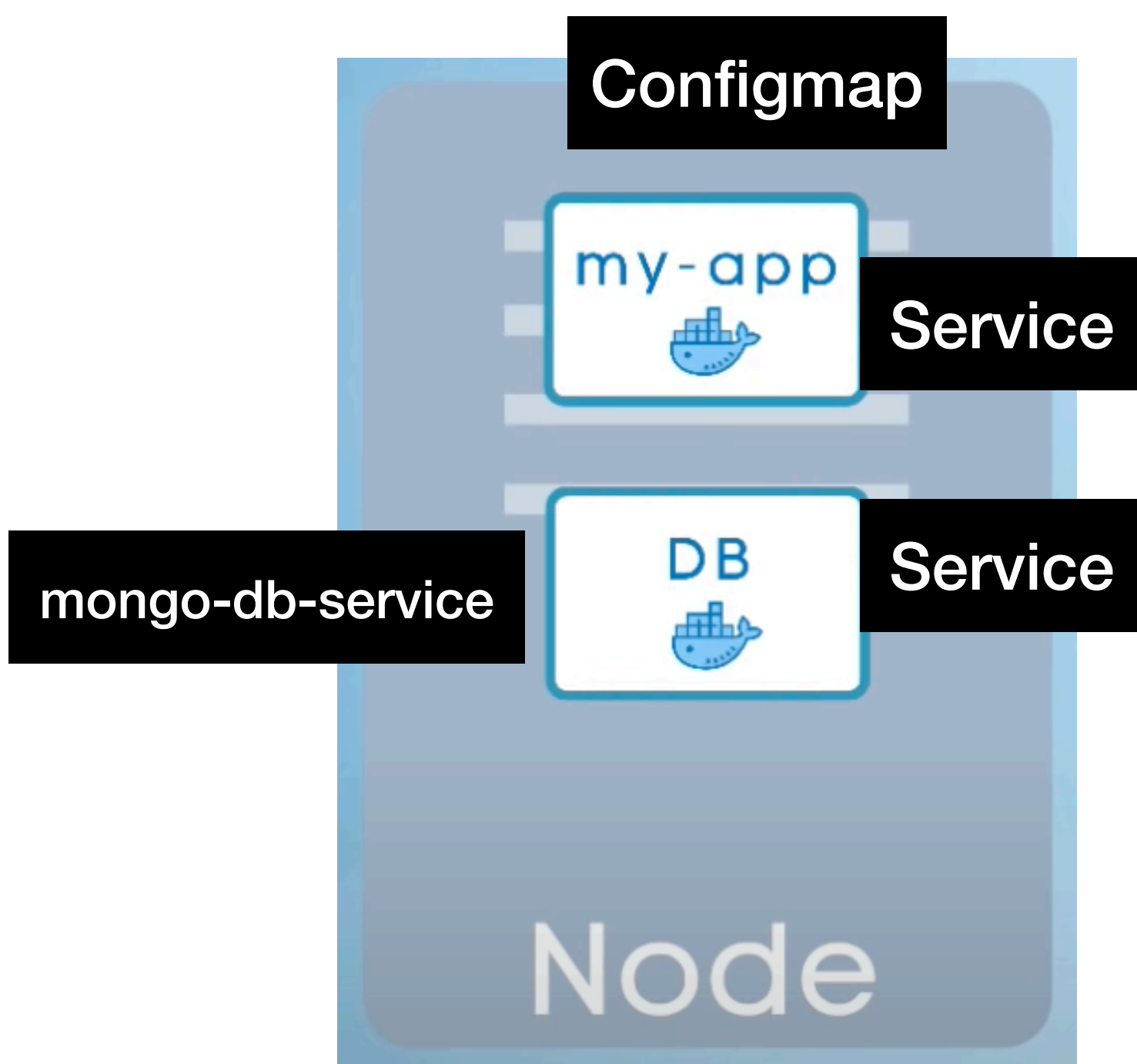


`http://node ip:port service`

`https://app name`

Komponenti Config map

- Kje specificiramo endpoint?
- Običajno v conf od aplikacije



Re-build

Repo

Pull v POD



Zunanja konfiguracija aplikacije

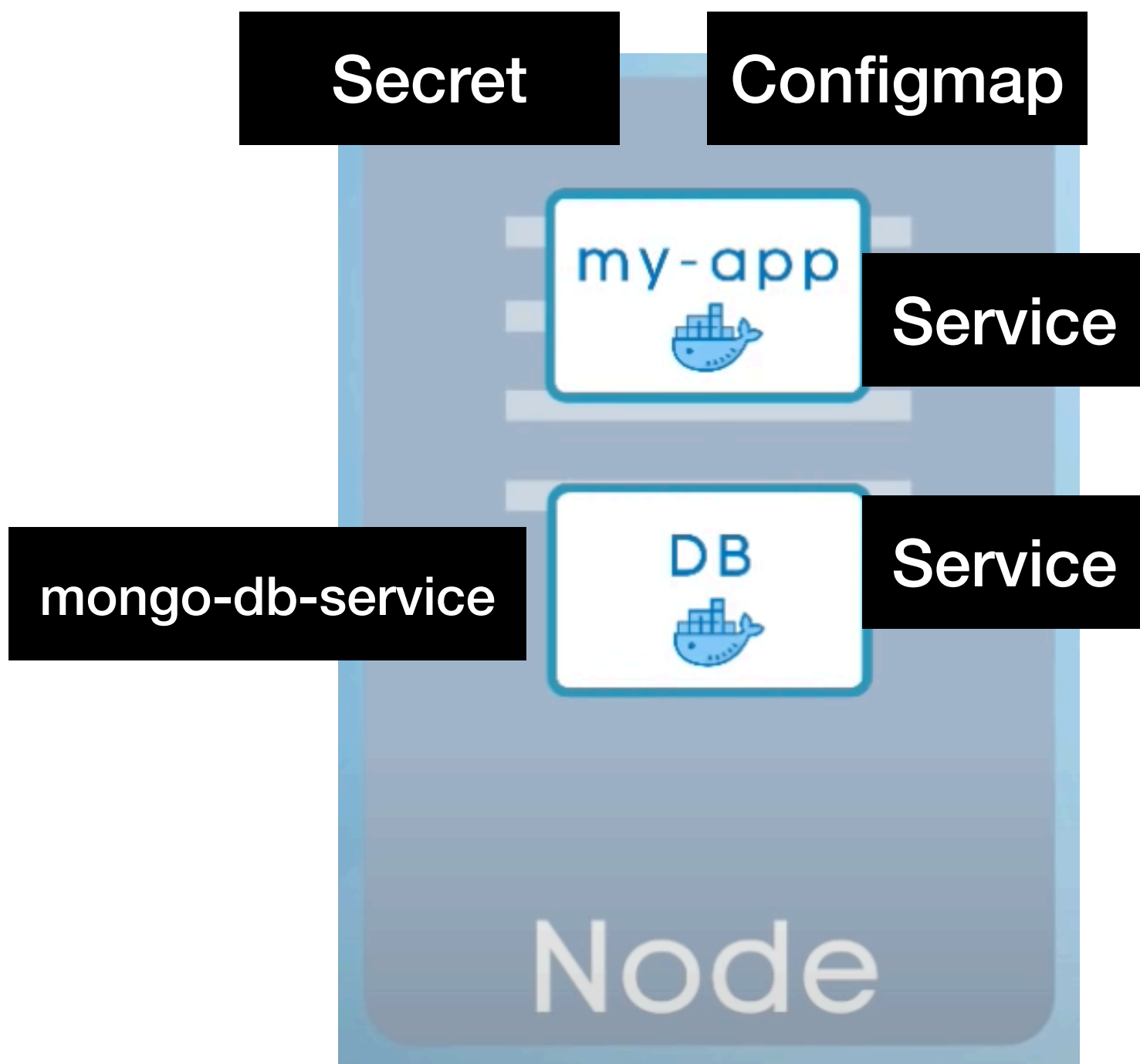
Samo za "javne" podatke

Komponenti

Secret



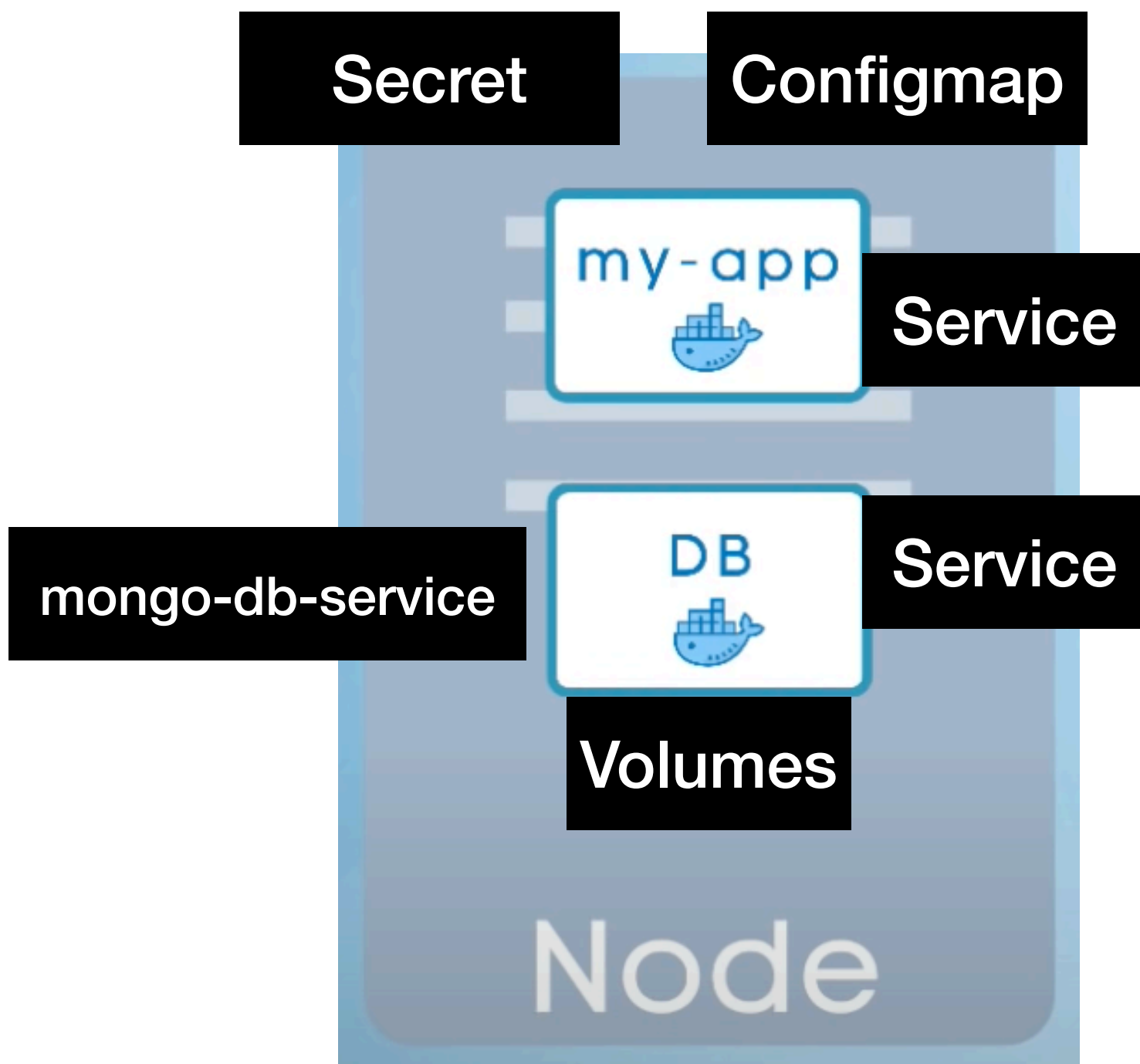
- Vpisana je v base64
- Enkripcija 3rd party
- Referenca do Pod
- Se lahko uporablja kot property ali env vars



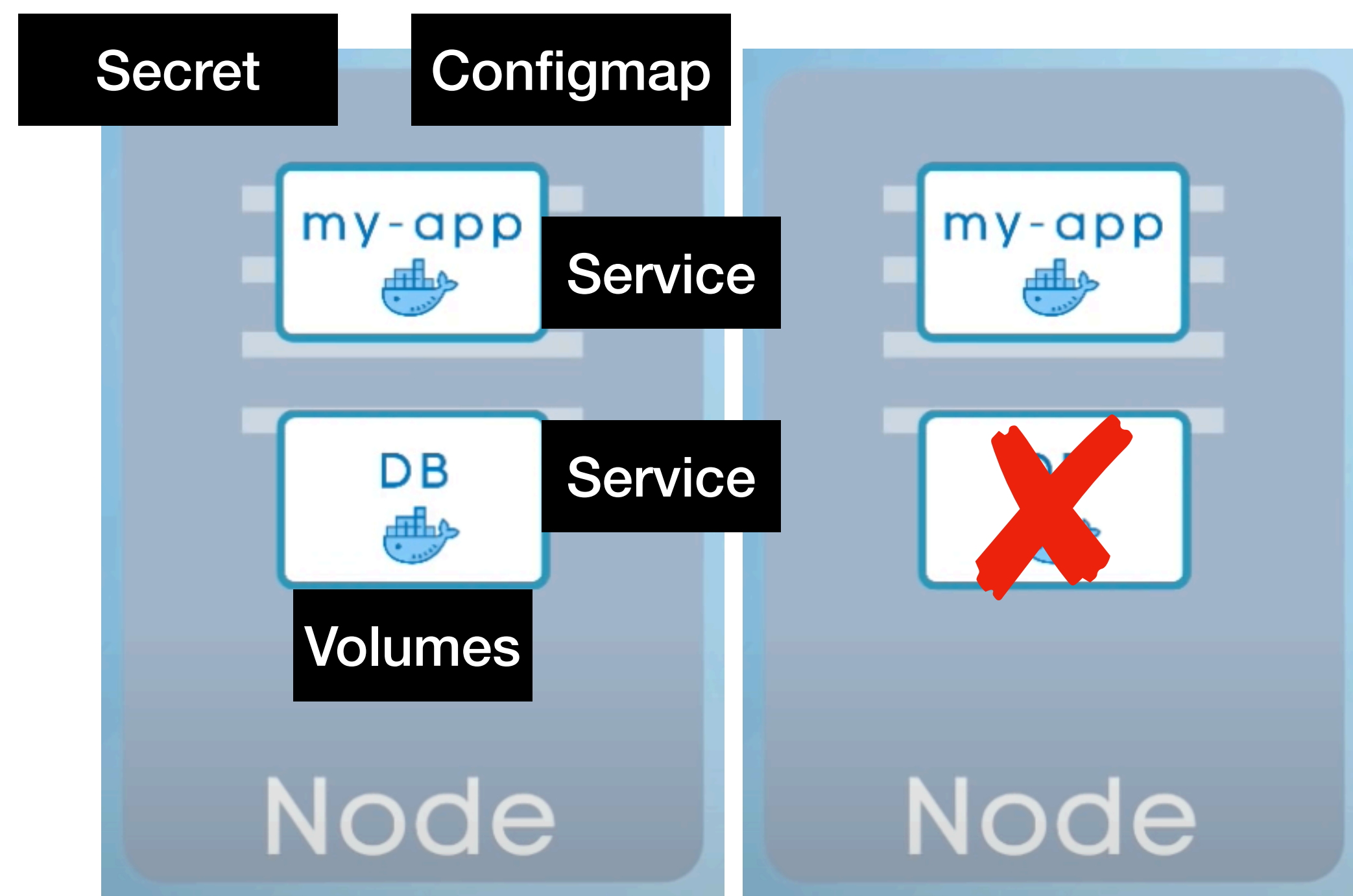
Komponenti Volumes



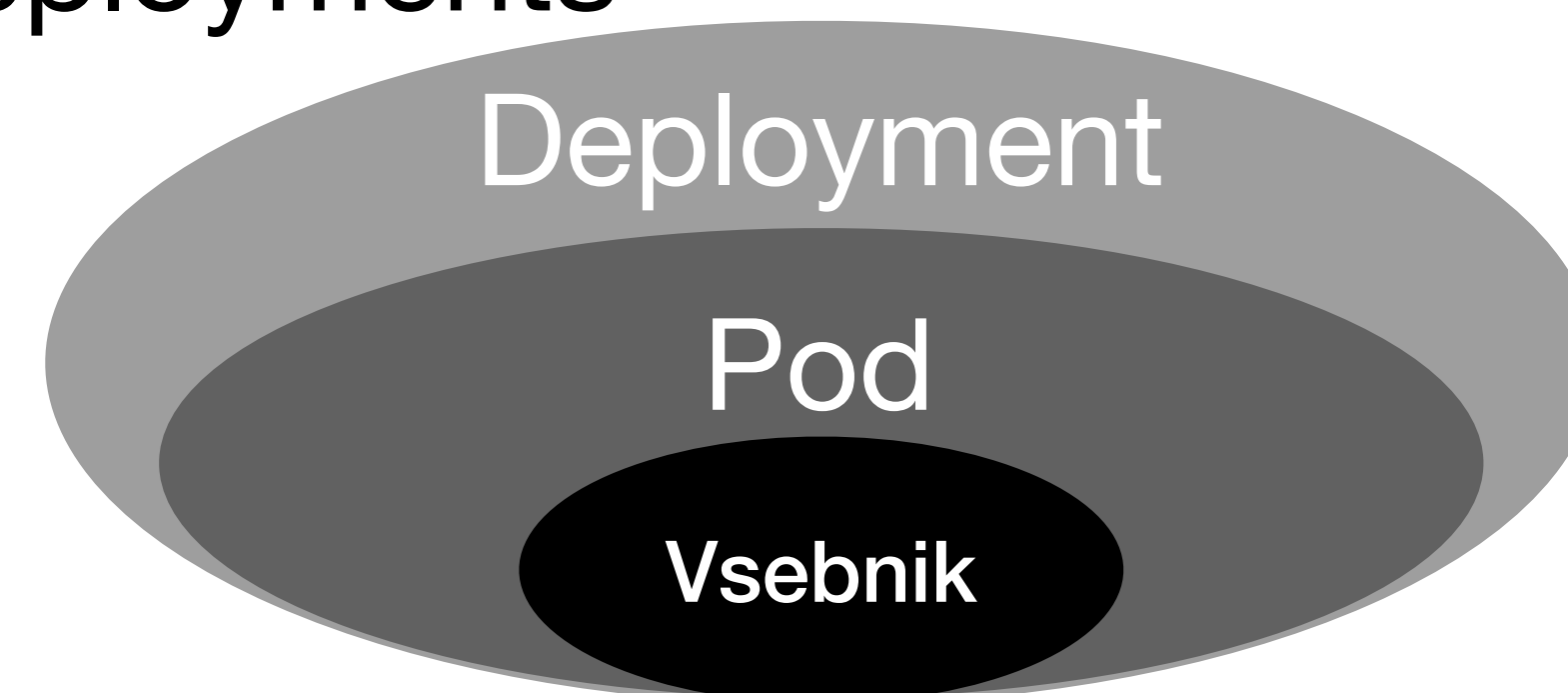
- Povezuje fizični diskovni prostor do pod
 - Lokalno
 - Remote



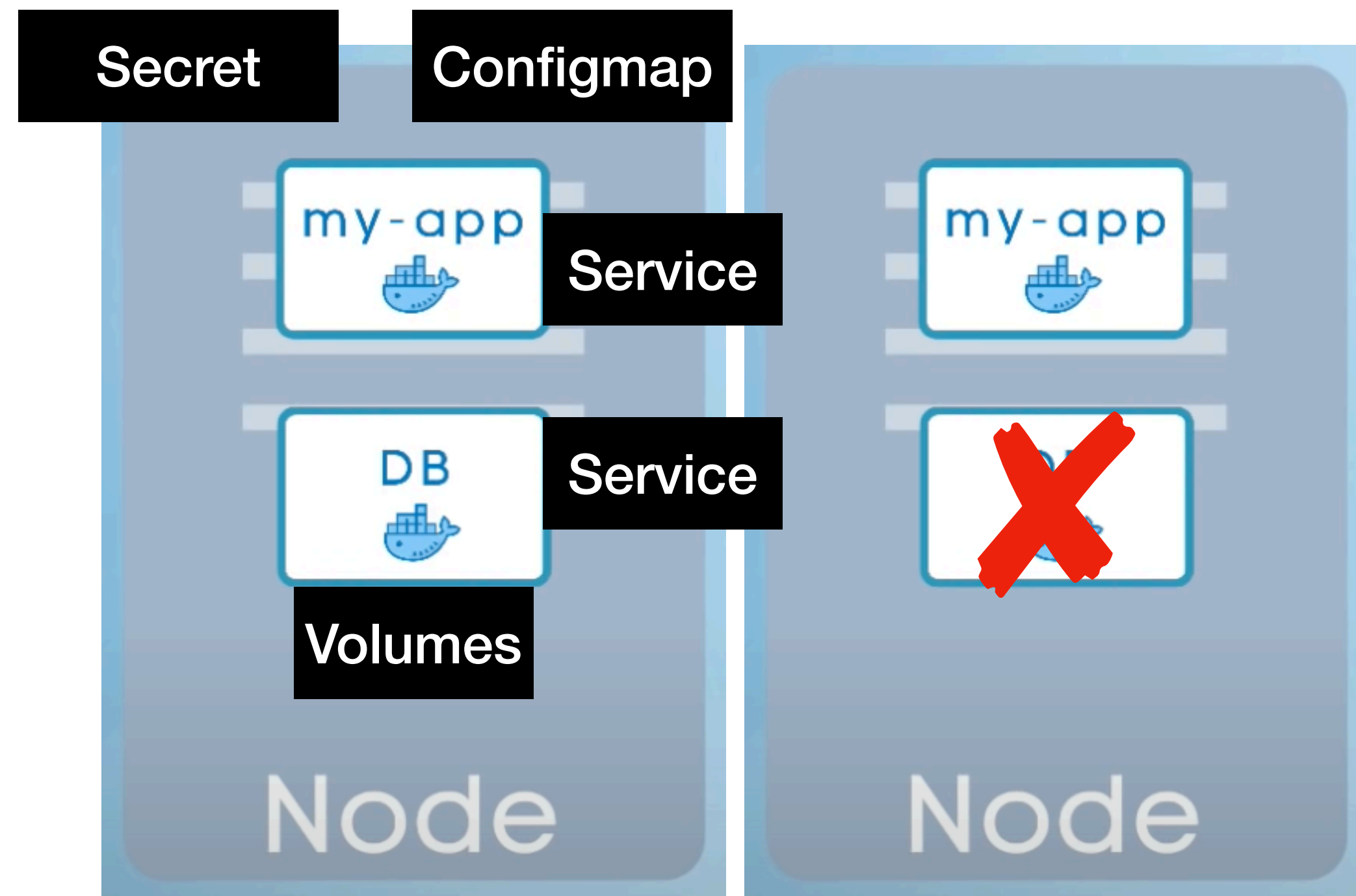
Komponenti Deployment



- Replikacija vsega!
- Service / fiksni IP
- Load balancer
- Specificiranje blueprint / plan
- V osnovi nikoli ne delamo pods ročno - vedo delamo Deployments



Komponenti Statefulset



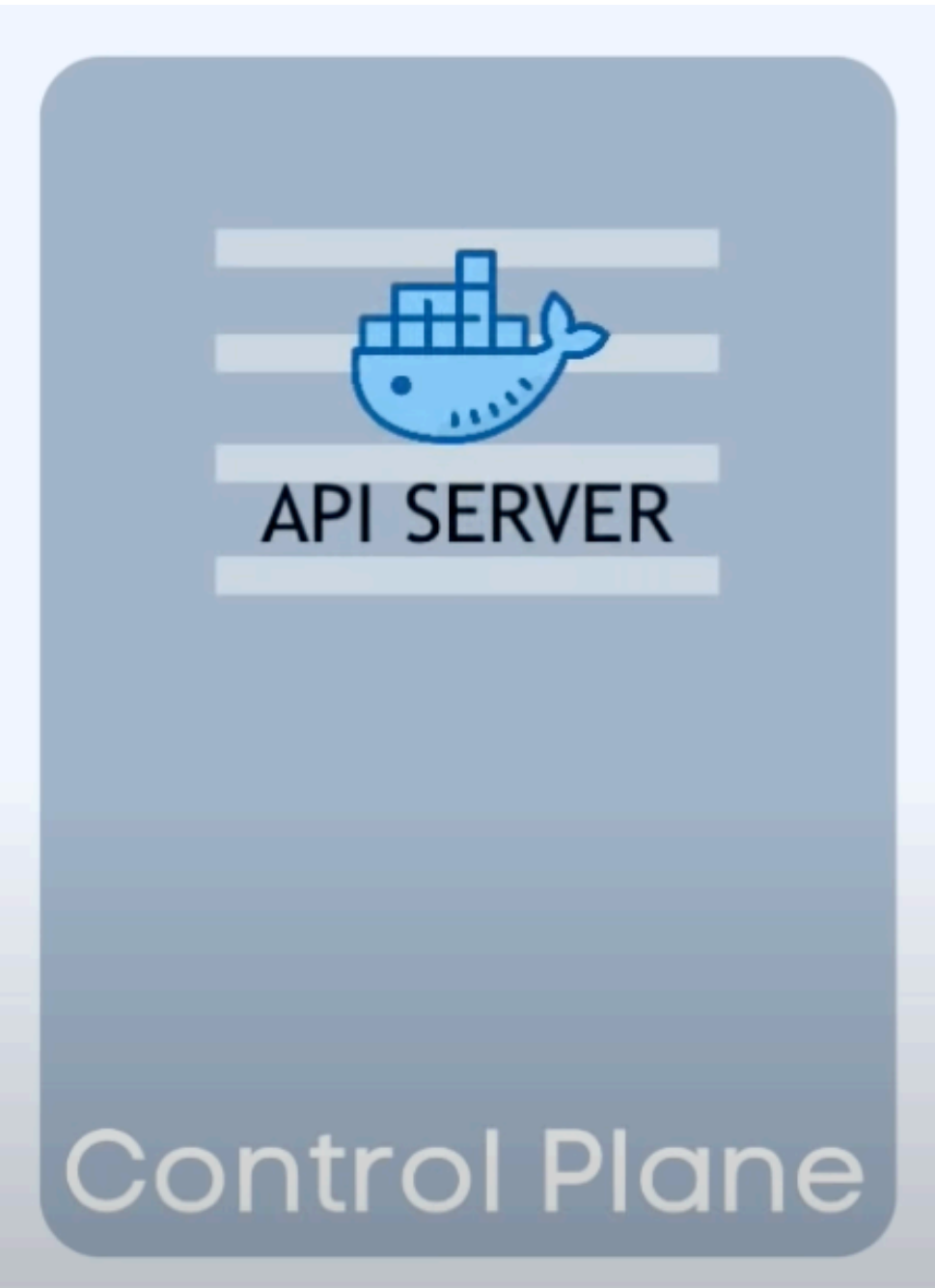
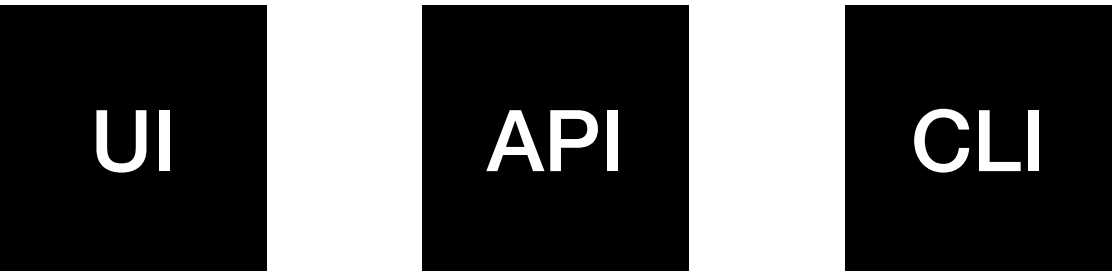
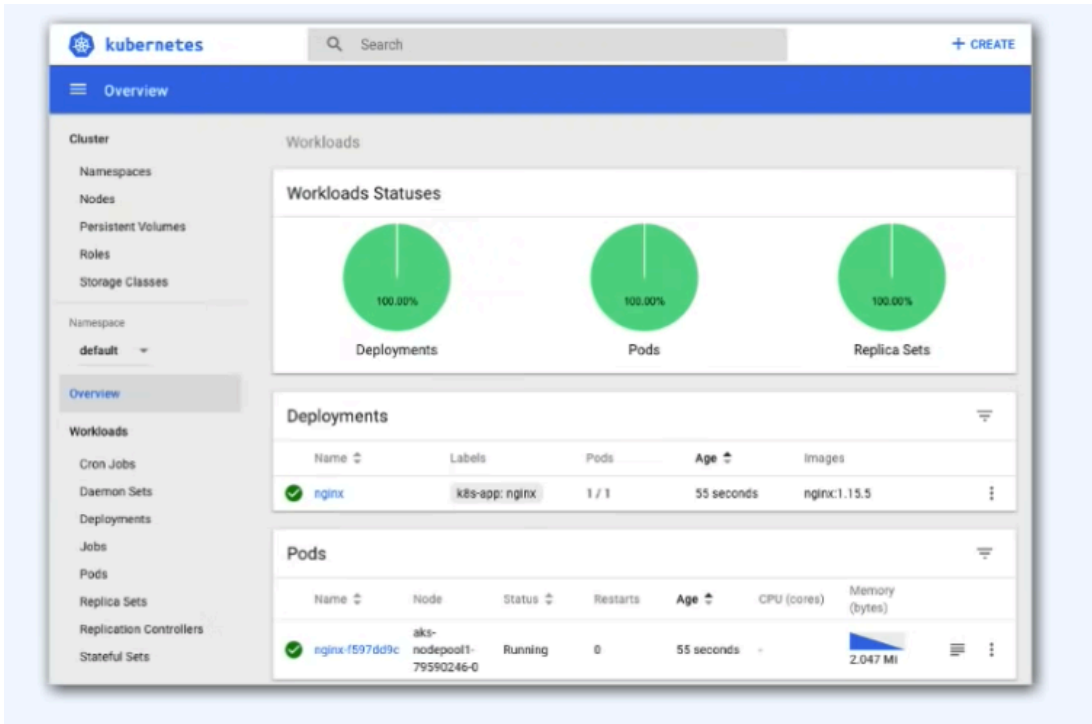
- Stateful apps
 - DBs
 - Običajno DB so zunaj K8s



- SateLESS

Konfiguracija

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: webapp-deployment
  labels:
    app: webapp
spec:
  replicas: 2
  selector:
    matchLabels:
      app: webapp
  template:
    metadata:
      labels:
        app: webapp
    spec:
      containers:
      - name: webapp
        image: my_image
        ports:
        - containerPort: 3000
        env:
        - name: USER_NAME
          valueFrom:
            secretKeyRef:
              name: mongo-secret
              key: mongo-user
        - name: USER_PWD
          valueFrom:
            secretKeyRef:
              name: mongo-secret
              key: mongo-password
        - name: DB_URL
          valueFrom:
            configMapKeyRef:
              name: mongo-config
              key: mongo-url
```



Logične komponente

Konfiguracija

Metadata

Specification

Status

- K8s ves čas posodablja
-

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: webapp-deployment
  labels:
    app: webapp
spec:
  replicas: 2
  selector:
    matchLabels:
      app: webapp
  template:
    metadata:
      labels:
        app: webapp
    spec:
      containers:
      - name: webapp
        image: my_image
        ports:
        - containerPort: 3000
        env:
        - name: USER_NAME
          valueFrom:
            secretKeyRef:
              name: mongo-secret
              key: mongo-user
        - name: USER_PWD
          valueFrom:
            secretKeyRef:
              name: mongo-secret
              key: mongo-password
        - name: DB_URL
          valueFrom:
            configMapKeyRef:
              name: mongo-config
              key: mongo-url
```

Logične komponente

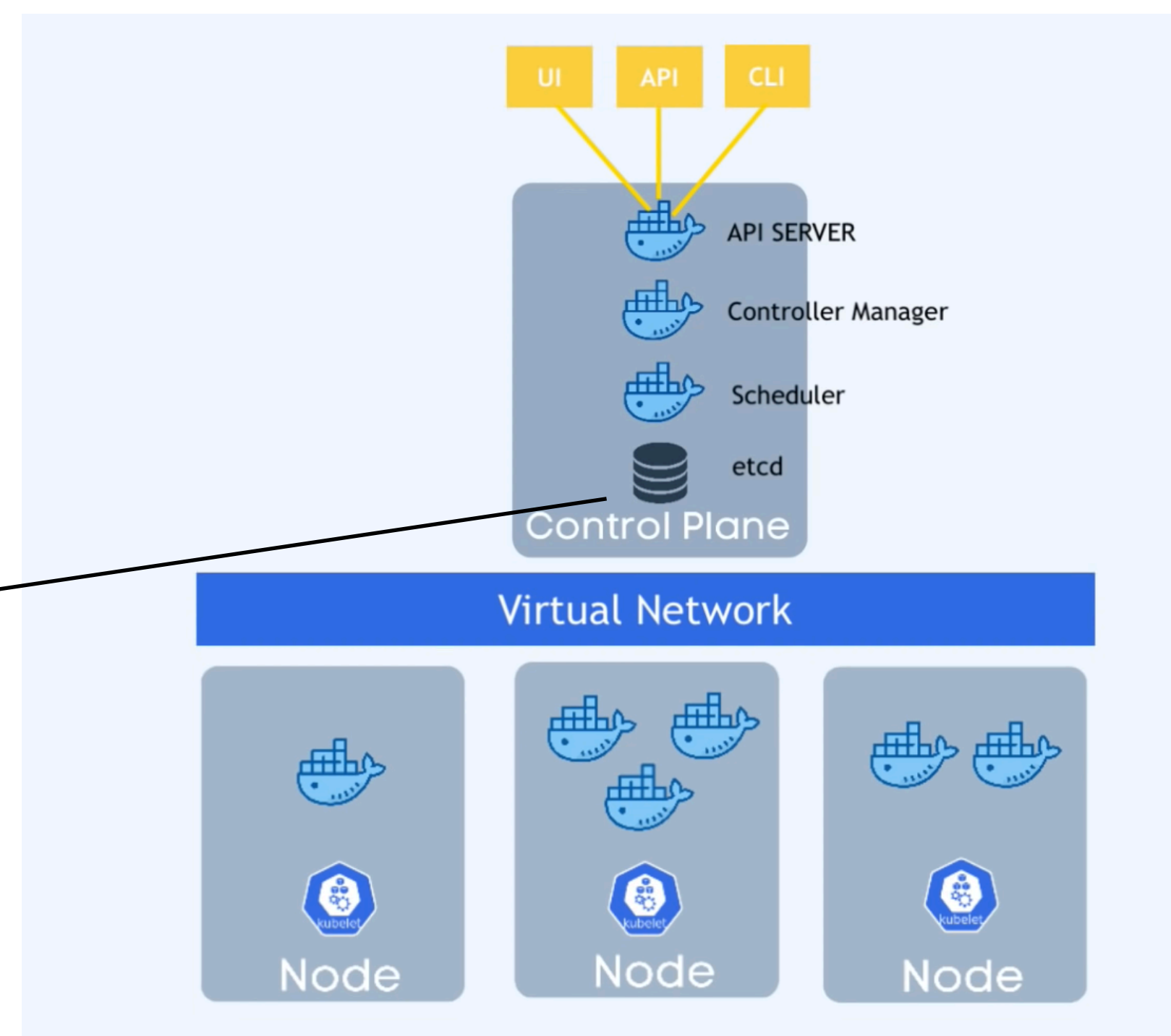
Konfiguracija

Metadata

Specification

Status

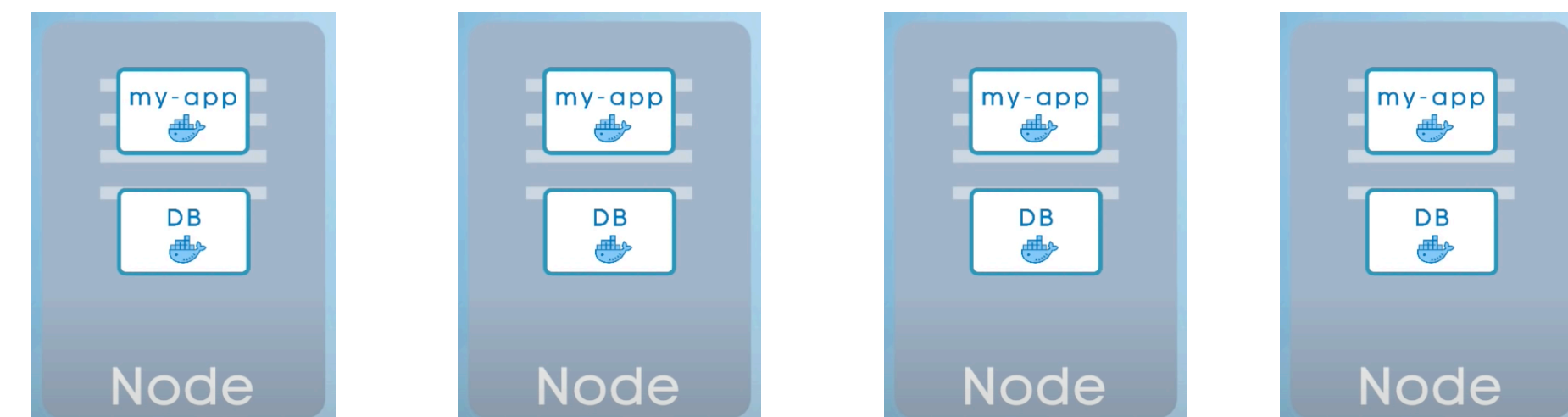
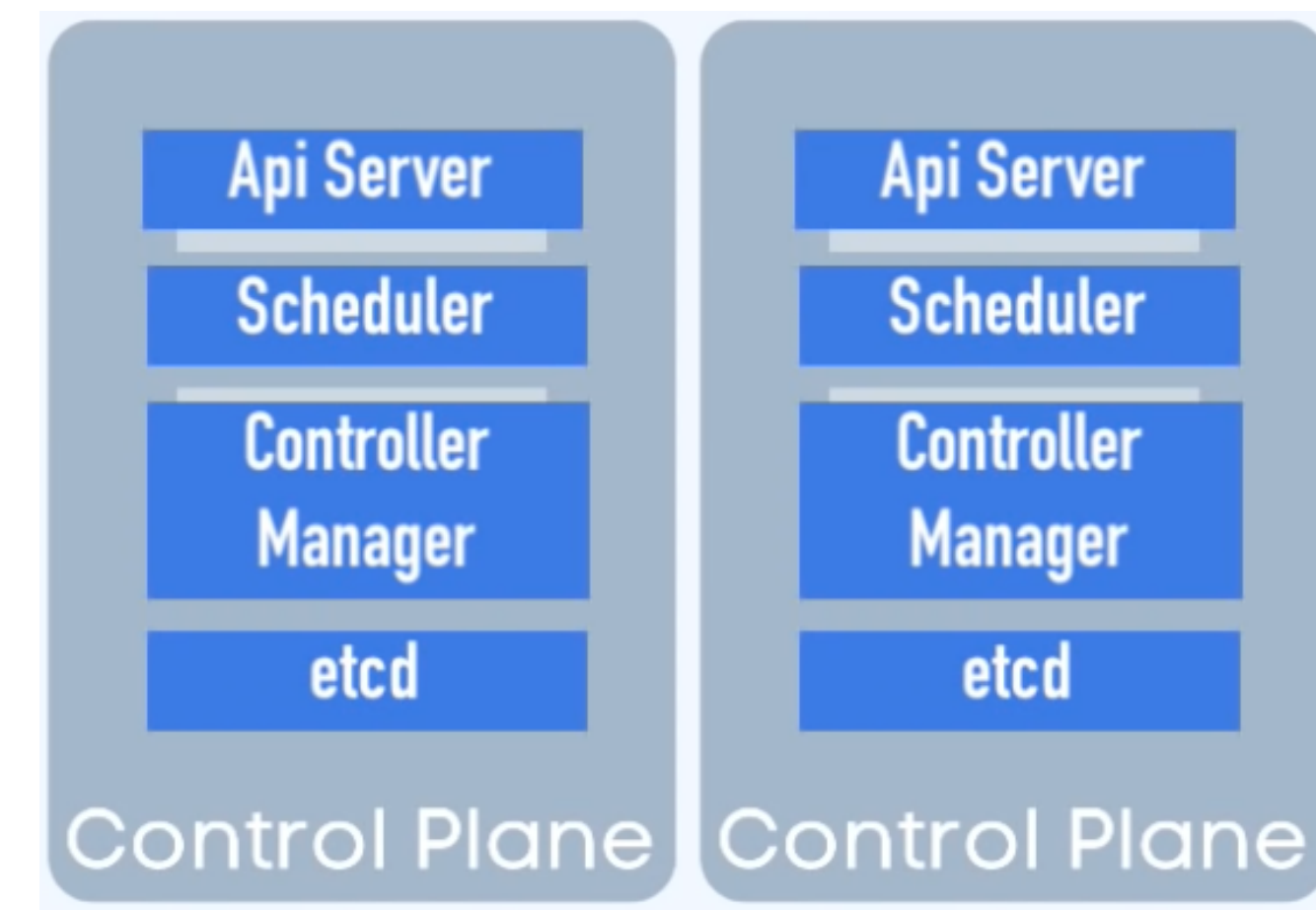
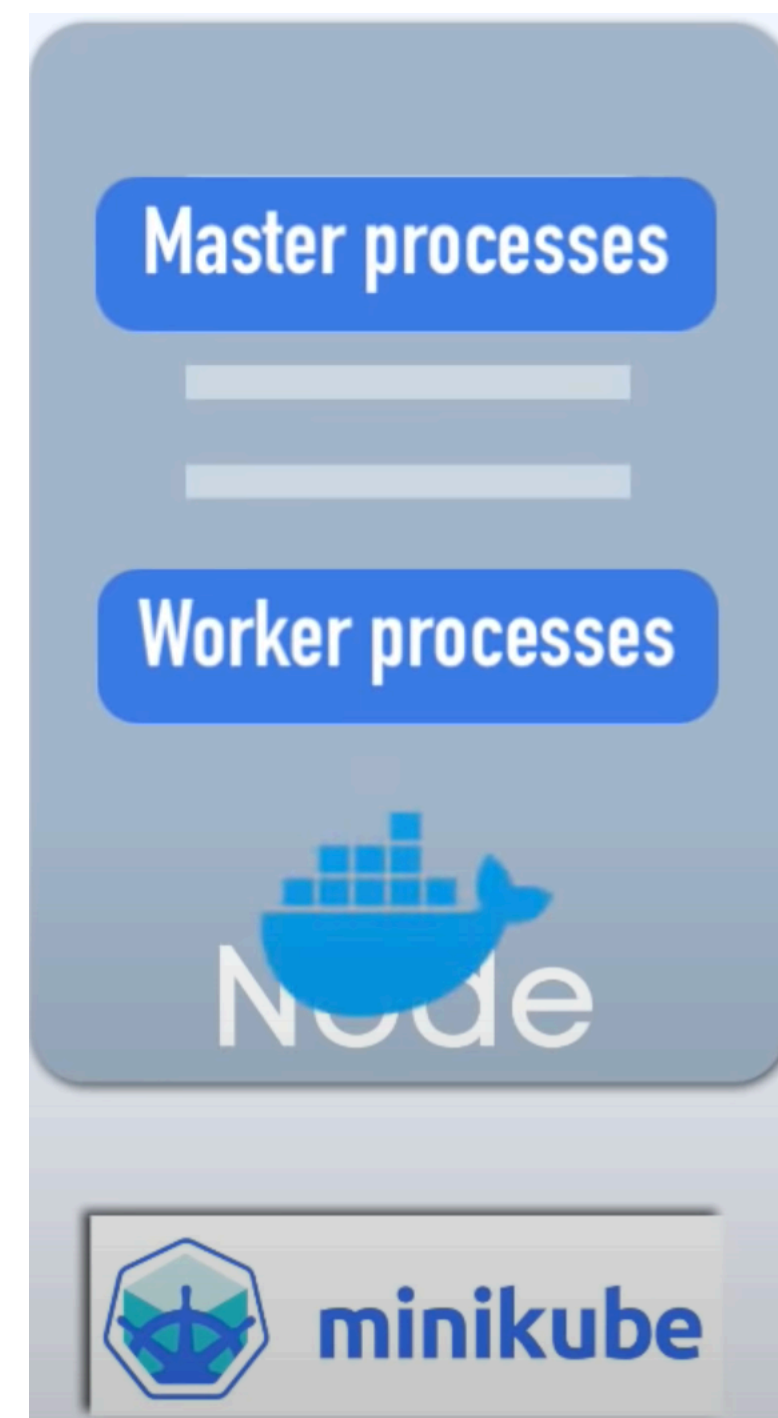
- K8s ves čas posodablja
- etcd vsebuje status vseh pods
 - K,v pairs



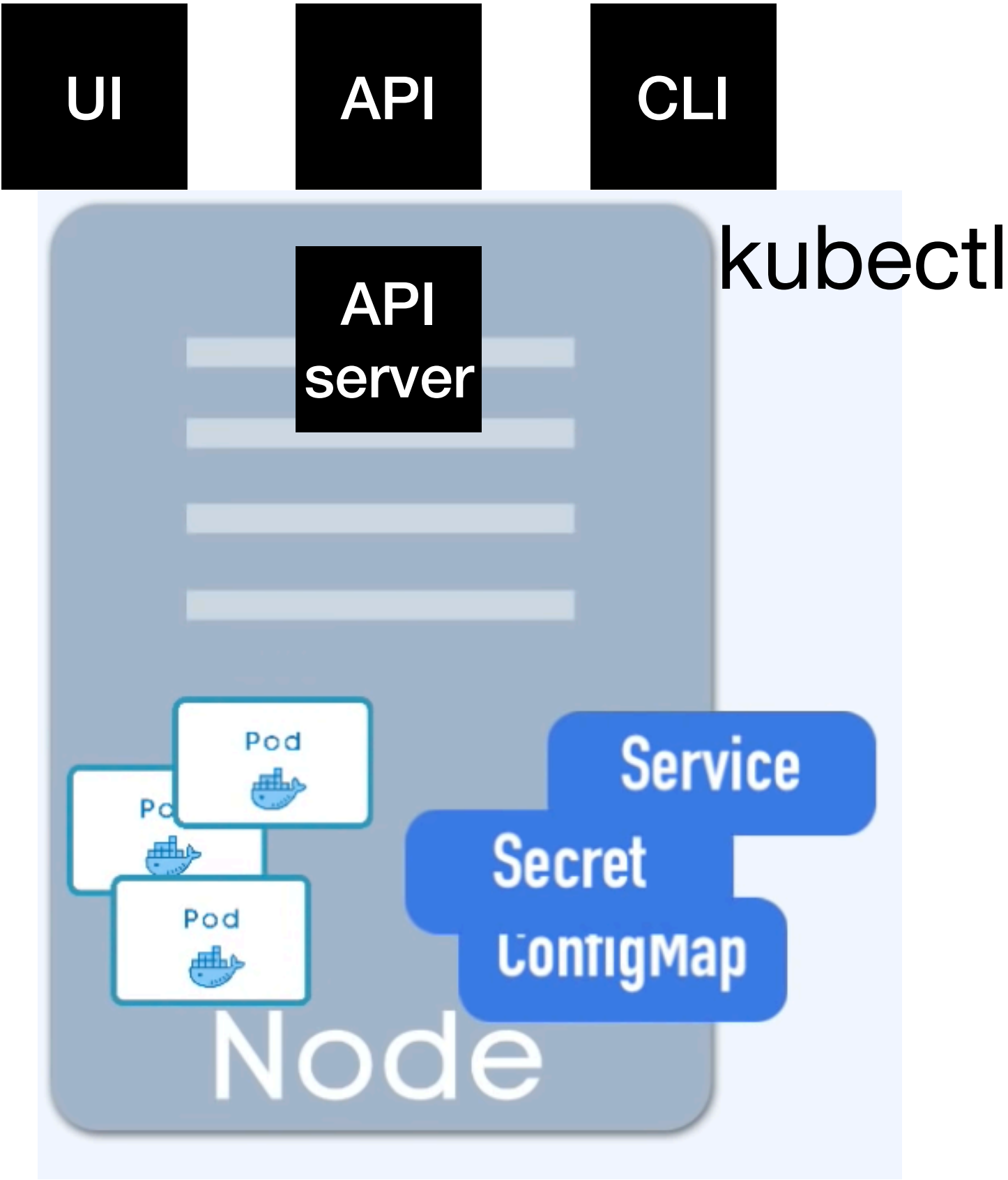
Minicube / kubectl

- Kaj je Minicube?
- Kaj je kubectl?
- Kako postavimo svoj K8s?

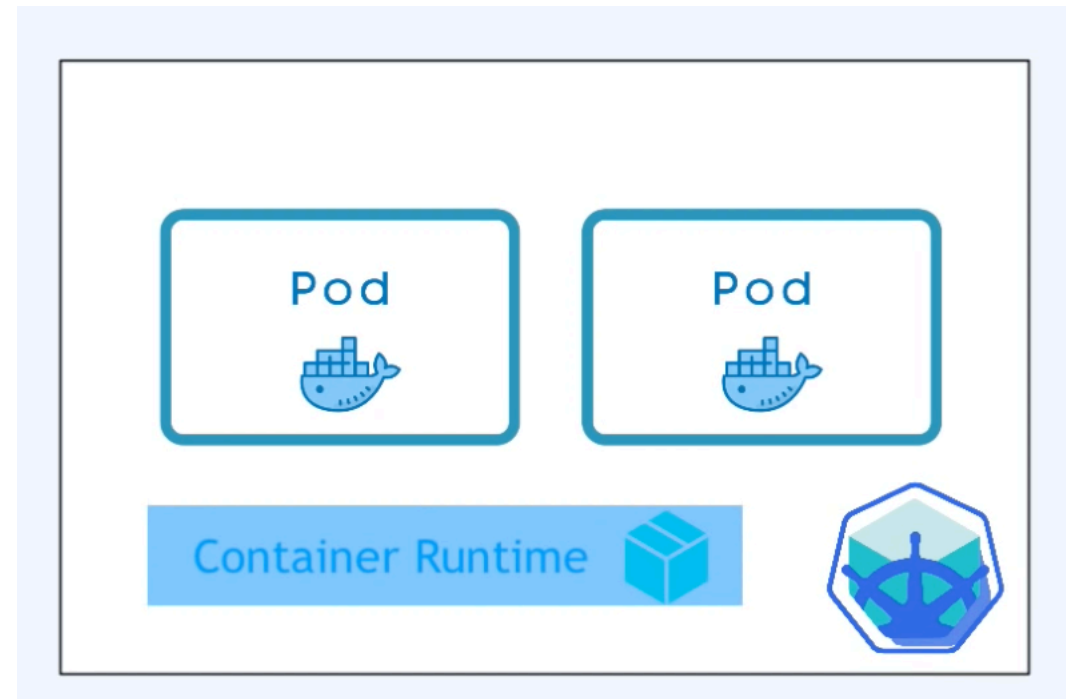
Običajna arhitektura gruča



kubectl



Minicube



```
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/manifests/1.22.1
##### 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/blobs/sha256:1c7f8280868fc960791f61
==> Downloading from https://pkg-containers.githubusercontent.com/ghcr1/blobs/sha256:1c7f8280868fc9
##### 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/minikube/manifests/1.23.2
##### 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/minikube/blobs/sha256:0103f2958d7609a9157e7de9c69b
==> Downloading from https://pkg-containers.githubusercontent.com/ghcr1/blobs/sha256:0103f2958d7609
##### 100.0%
==> Installing dependencies for minikube: kubernetes-cli
==> Installing minikube dependency: kubernetes-cli
==> Pouring kubernetes-cli--1.22.1.mojave.bottle.tar.gz
```

`minicube start --driver docker`

`minicube status`

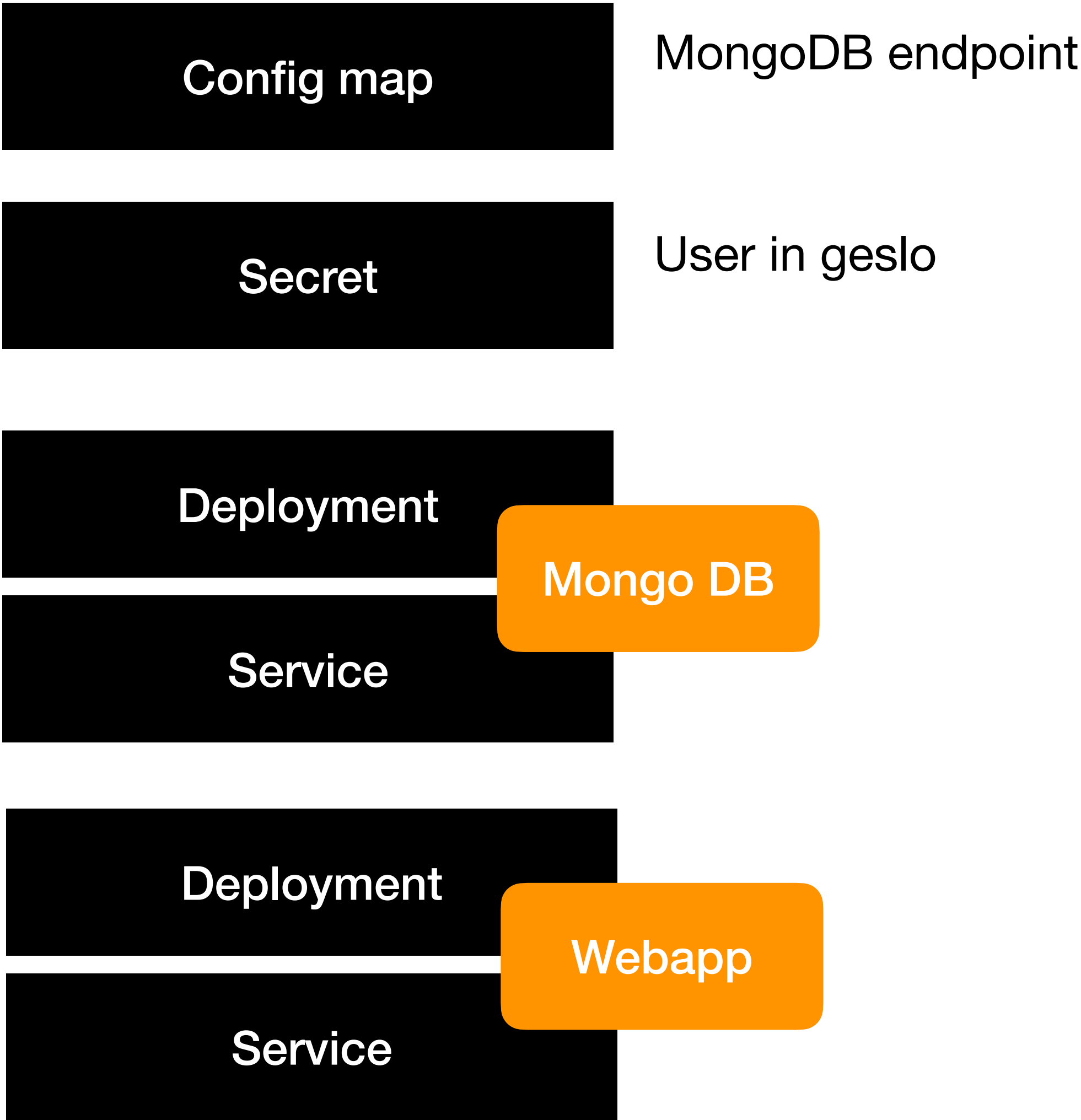
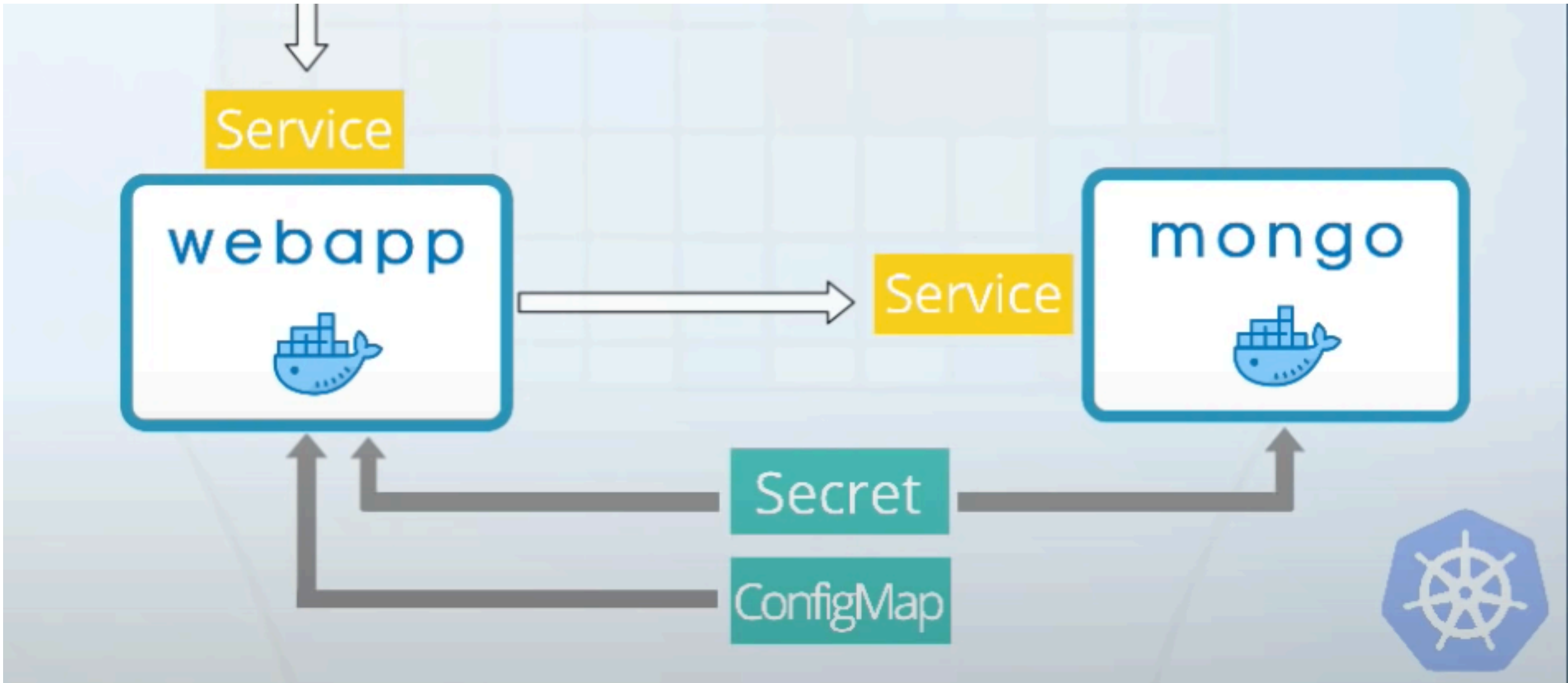
Minicube

kubectl

kubectl get node

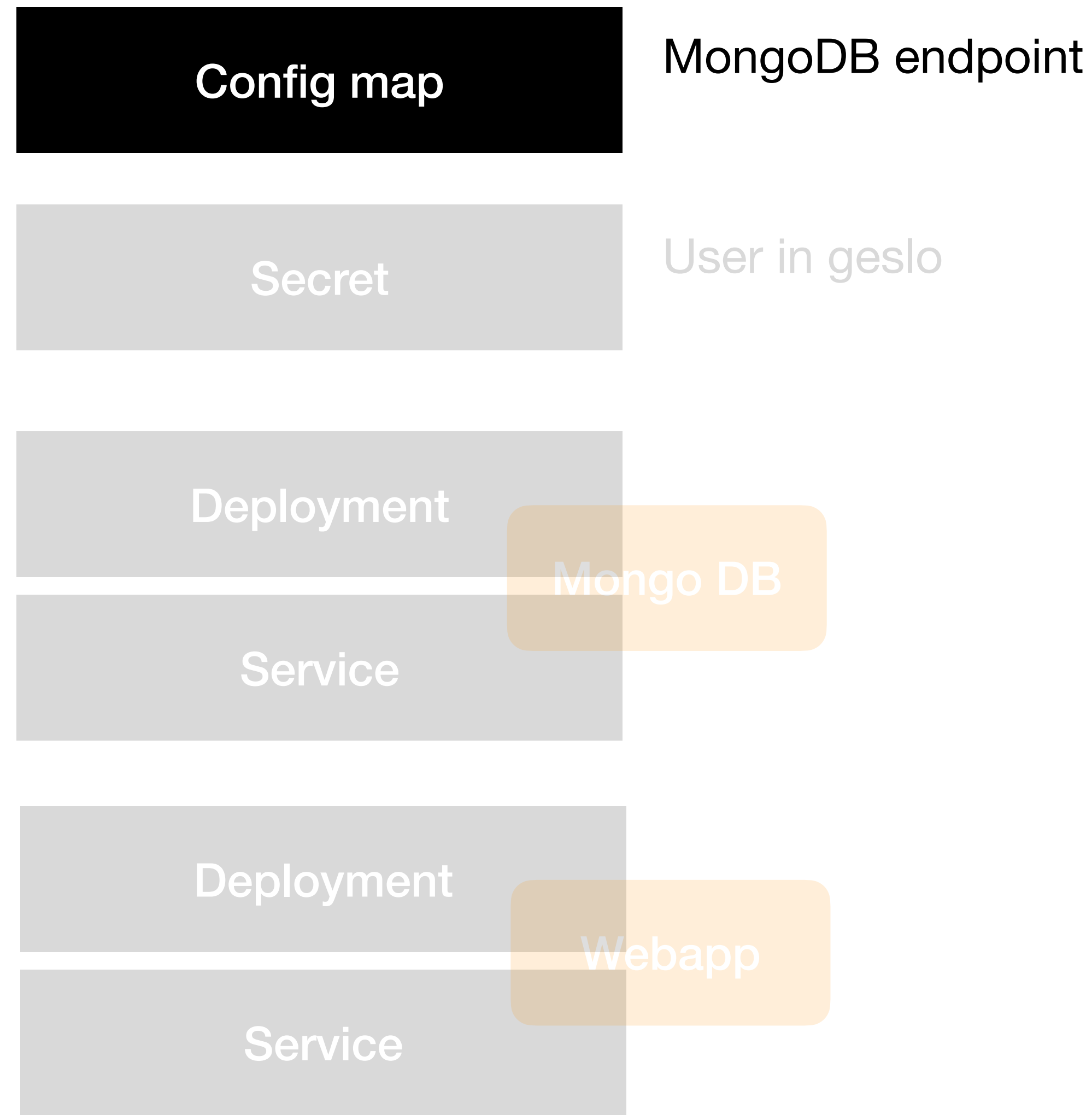
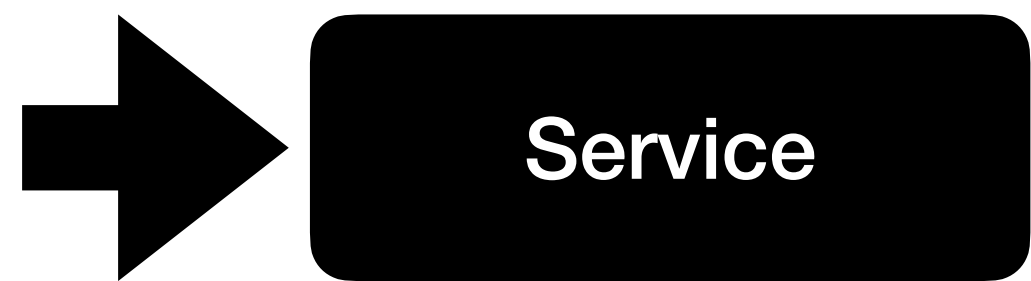
NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane,master	117s	v1.22.2

Demo app



mongo-config.yaml

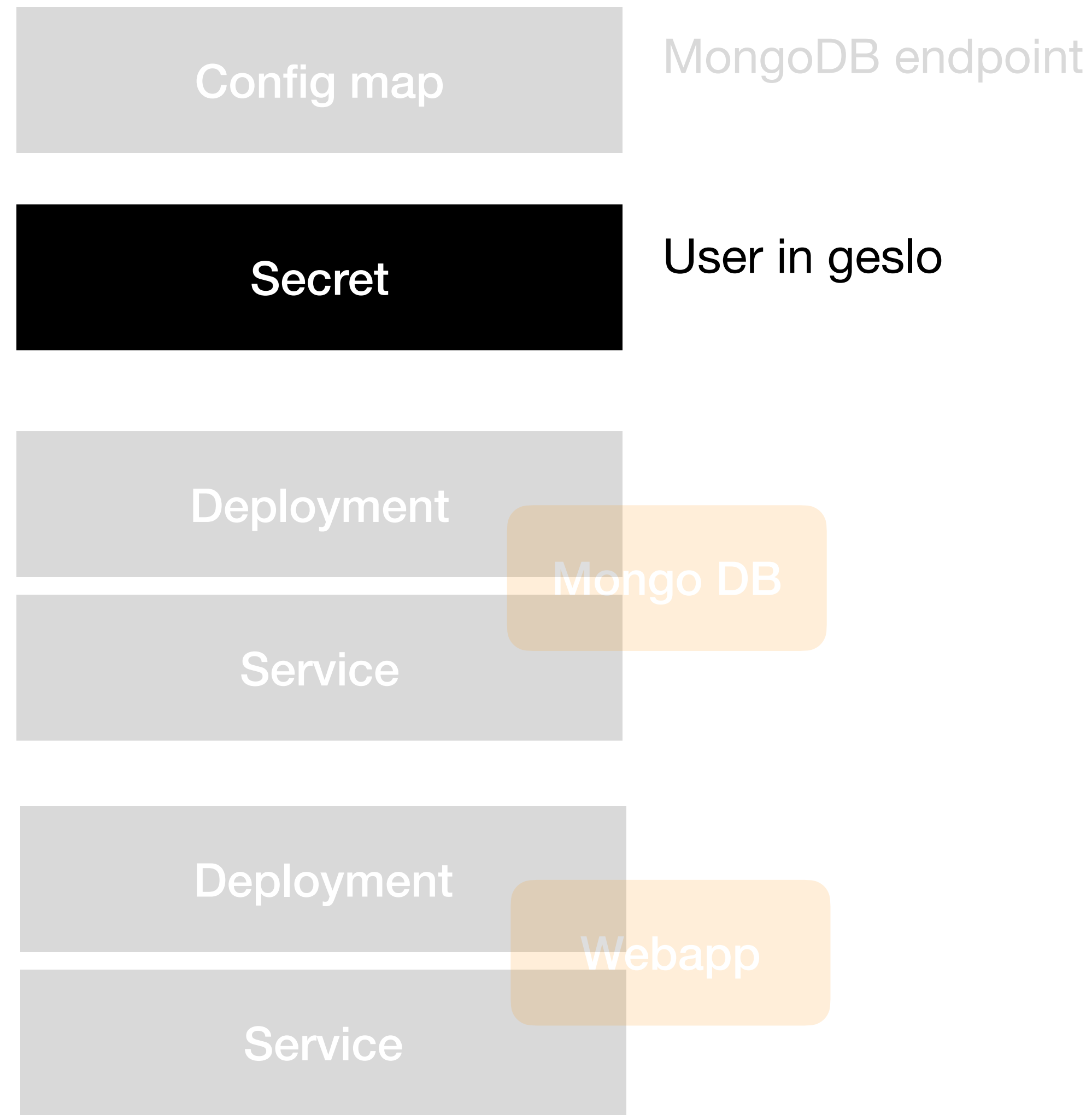
```
apiVersion: v1
kind: ConfigMap
metadata:
  name: mongo-config
data:
  mongo-url: mongo-service
```



mongo-secret.yaml

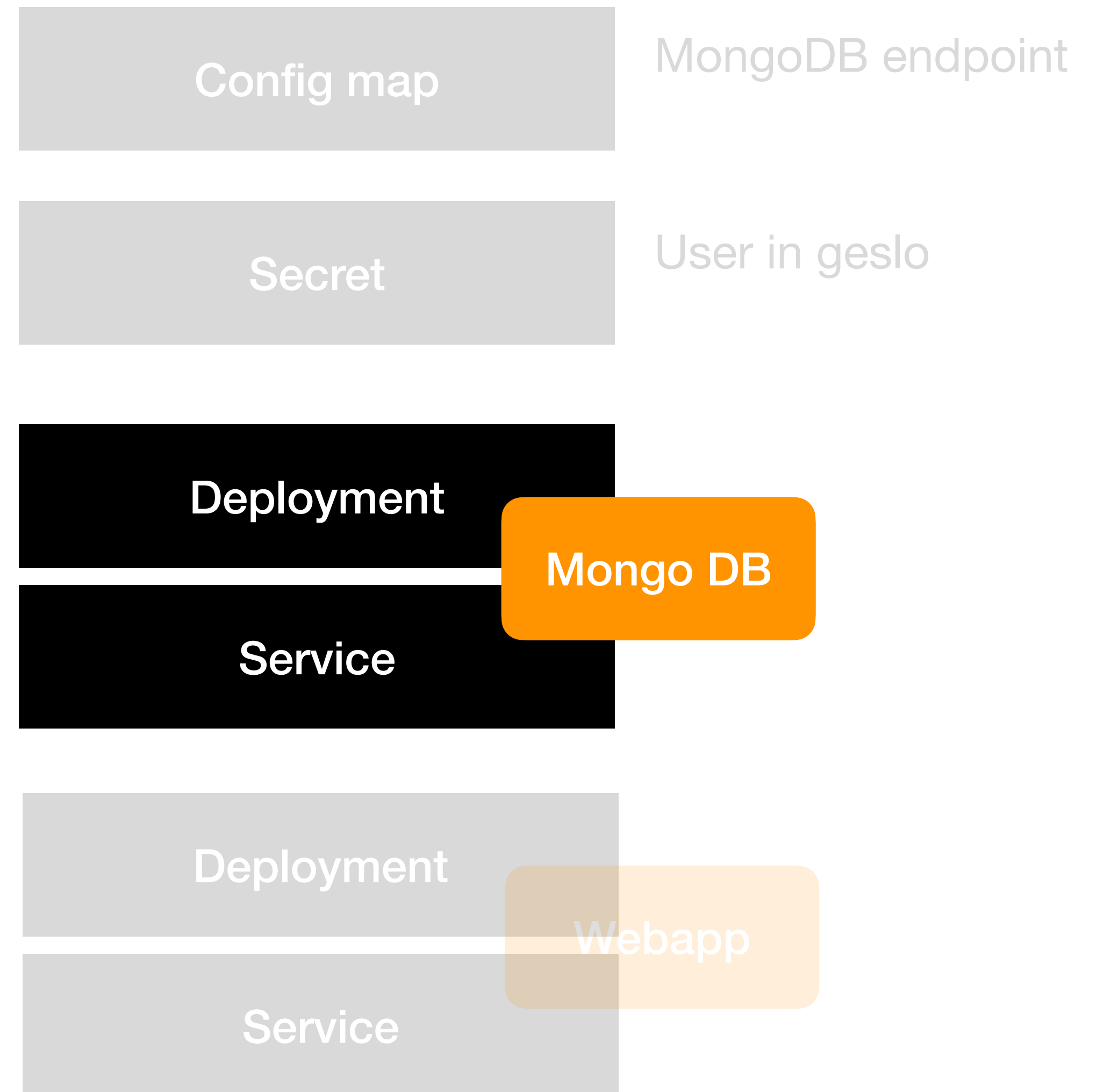
```
apiVersion: v1
kind: Secret
metadata:
  name: mongo-secret
type: Opaque
data:
  mongo-user: cGF2bGVhbnQ=
  mongo-password: cGF2bGVuMDA=
```

```
echo -n pavleb|base64
```



mongo.yaml

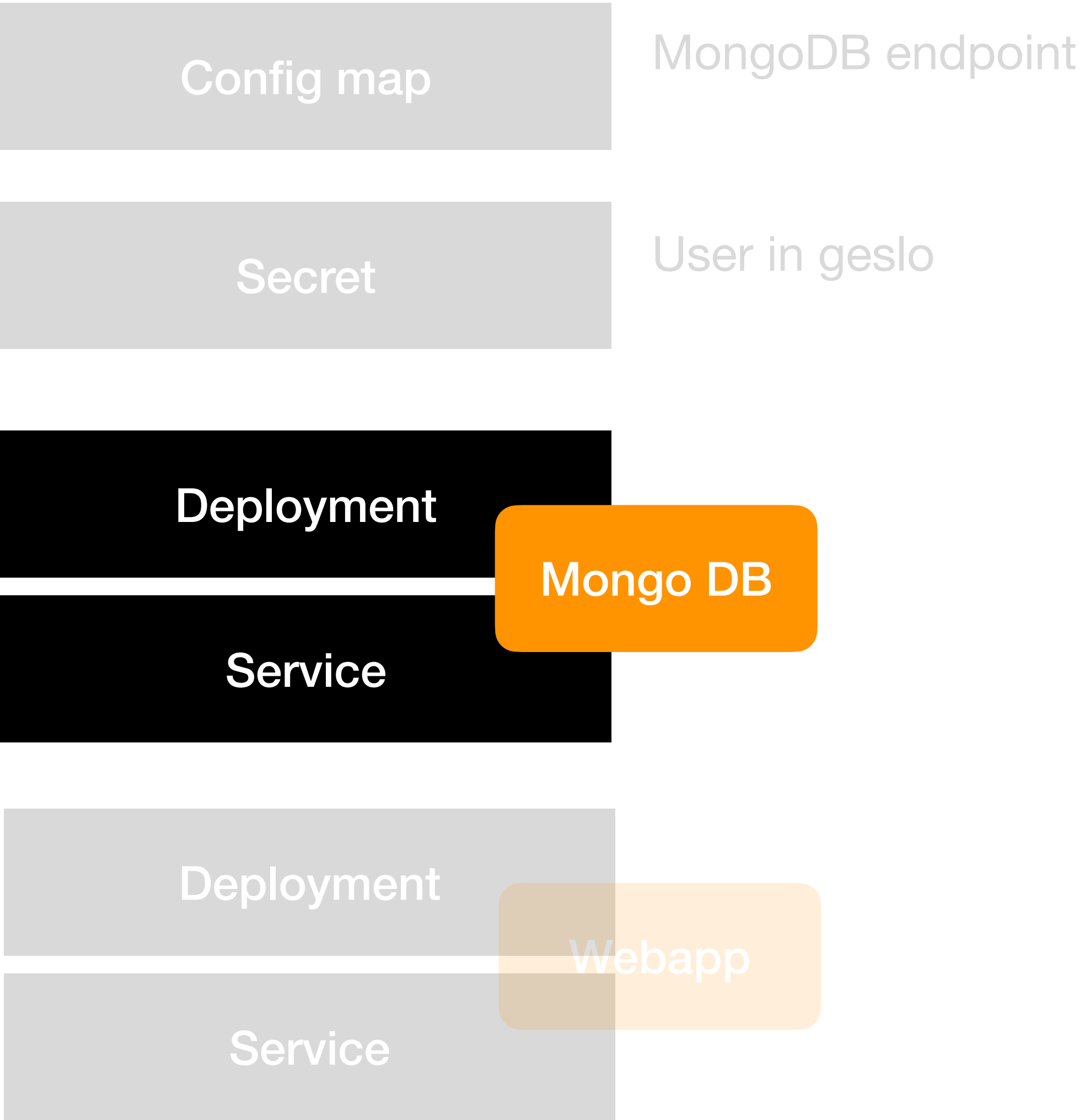
- Kind: Deployment
- Template
 - Specifikacija POD
- Container
 - Kateri image
 - Kateri port



mongo.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo-deployment
  labels:
    app: mongo
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongo
  template:
    metadata:
      labels:
        app: mongo
    spec:
      containers:
        - name: mongodb
          image: mongo:5.0
          ports:
            - containerPort: 27017
          env:
            - name:
MONGO_INITDB_ROOT_USERNAME
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-user
            - name:
MONGO_INITDB_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-password
```

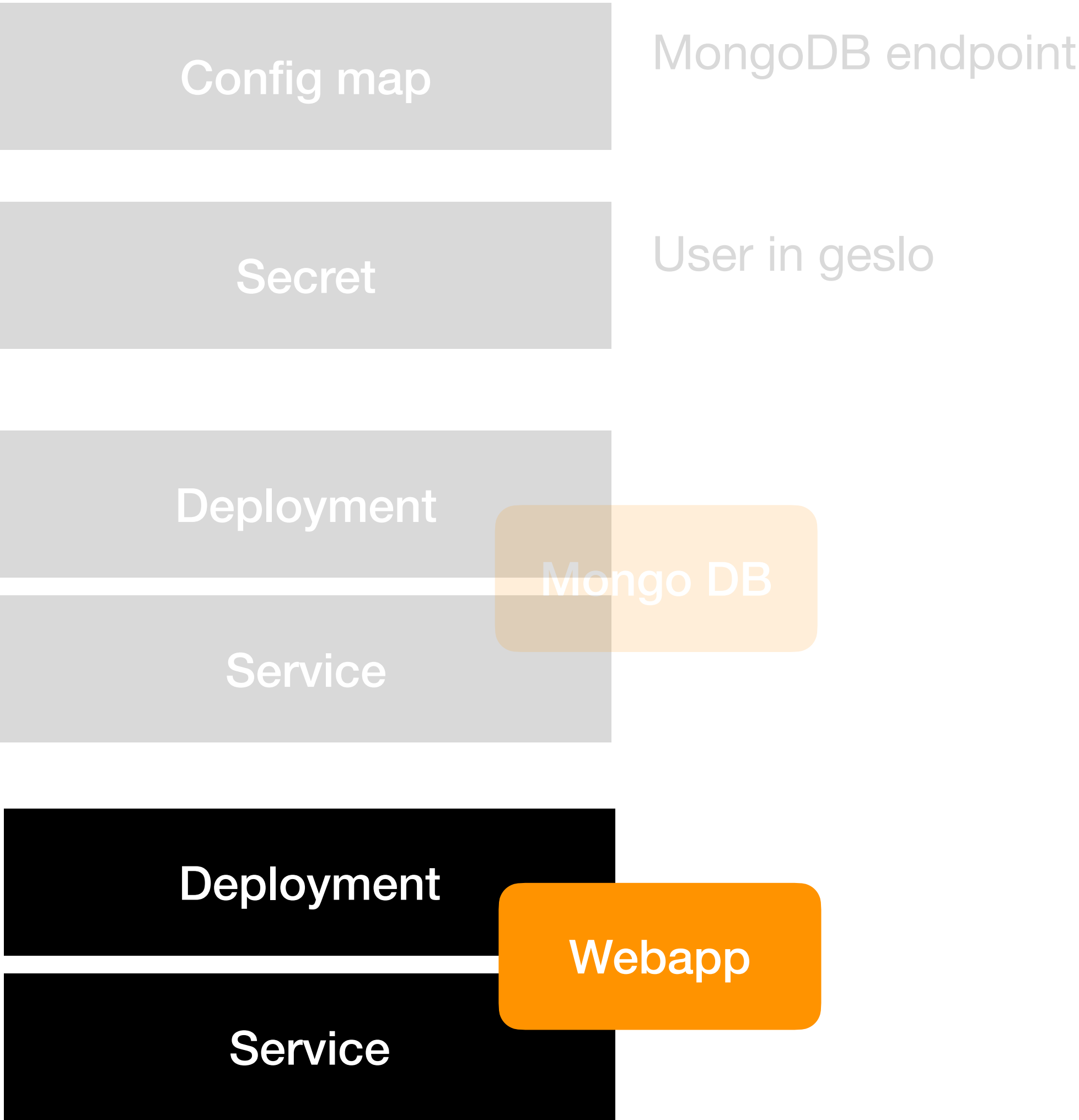
```
---
apiVersion: v1
kind: Service
metadata:
  name: mongo-service
spec:
  selector:
    app: mongo
  ports:
    - protocol: TCP
      port: 27017
      targetPort: 27017
```



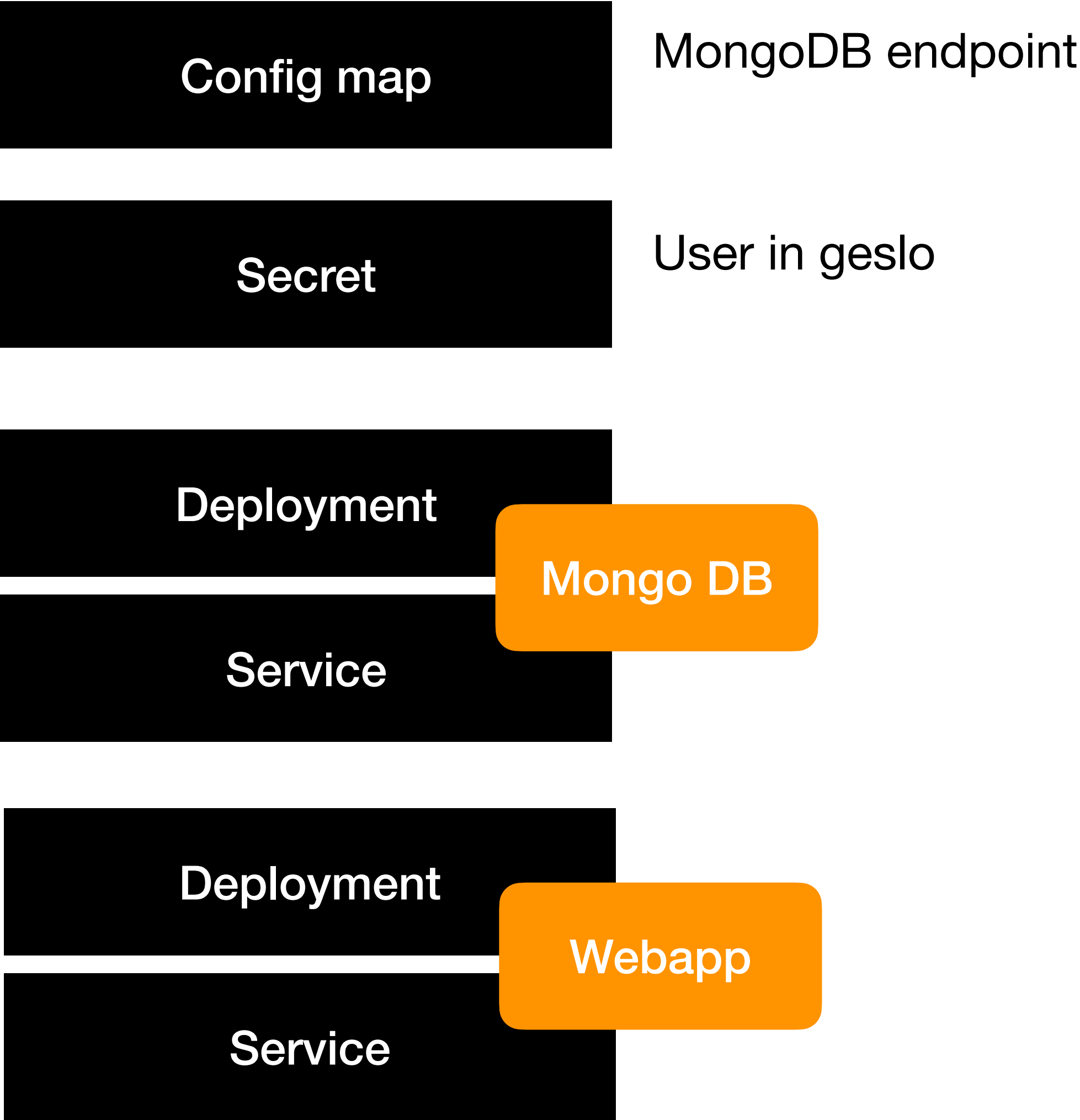
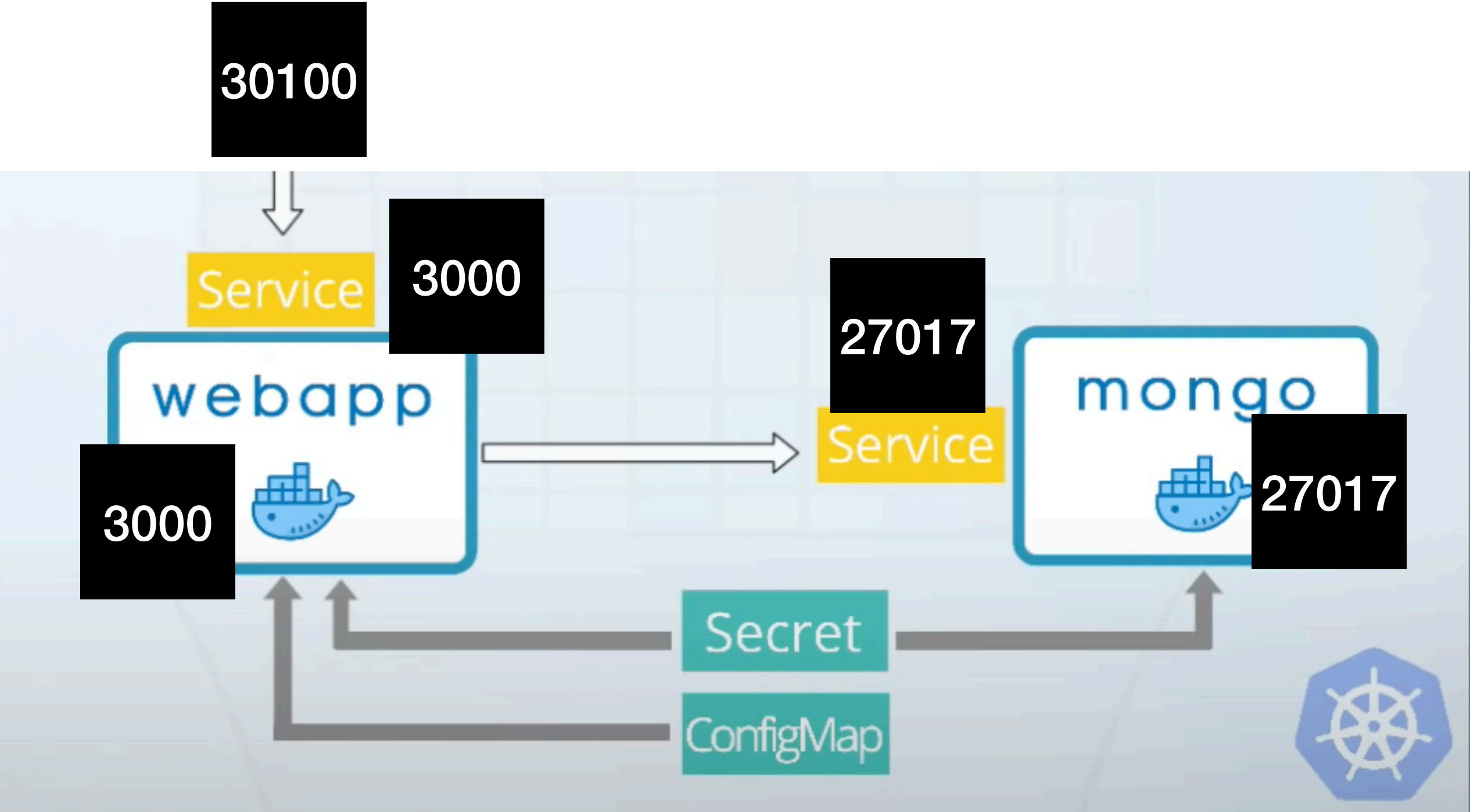
webapp.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: webapp-deployment
  labels:
    app: webapp
spec:
  replicas: 1
  selector:
    matchLabels:
      app: webapp
  template:
    metadata:
      labels:
        app: webapp
    spec:
      containers:
        - name: webapp
          image: nodejs
          ports:
            - containerPort: 3000
          env:
            - name: USER_NAME
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-user
            - name: USER_PWD
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-password
            - name: DB_URL
              valueFrom:
                configMapKeyRef:
                  name: mongo-config
                  key: mongo-url
```

```
apiVersion: v1
kind: Service
metadata:
  name: webapp-service
spec:
  type: NodePort
  selector:
    app: webapp
  ports:
    - protocol: TCP
      port: 3000
      targetPort: 3000
      nodePort: 30100
```



Demo app



Minicube

Deployments

```
kubectl apply -f mongo-config.yaml
```

```
kubectl apply -f mongo-secret.yaml
```

```
kubectl apply -f mongo.yaml
```

```
kubectl apply -f webapp.yaml
```

Minicube

Preverimo status

kubectl get all

NAME	READY	STATUS	RESTARTS	AGE
pod/mongo-deployment-7875498c-z477b	1/1	Running	0	51s
pod/webapp-deployment-dcffd6bcc-gsmlt	1/1	Running	0	37s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	190d
service/mongo-service	ClusterIP	10.111.46.184	<none>	27017/TCP	51s
service/webapp-service	NodePort	10.110.153.3	<none>	3000:30100/TCP	37s

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/mongo-deployment	1/1	1	1	51s
deployment.apps/webapp-deployment	1/1	1	1	37s

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/mongo-deployment-7875498c	1	1	1	51s
replicaset.apps/webapp-deployment-dcffd6bcc	1	1	1	37s

Minicube

Preverimo status

kubectl get svc

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	190d
mongo-service	ClusterIP	10.111.46.184	<none>	27017/TCP	7m6s
webapp-service	NodePort	10.110.153.3	<none>	3000:30100/TCP	6m52s